

THABELO MAHADA ST10449303
NKATEKO MABASA ST10448553
KGOLAGANO MPHAI ST10445320
THUBALAKHE MTHEMBU ST10447914

APPR6312 PART 1 POE

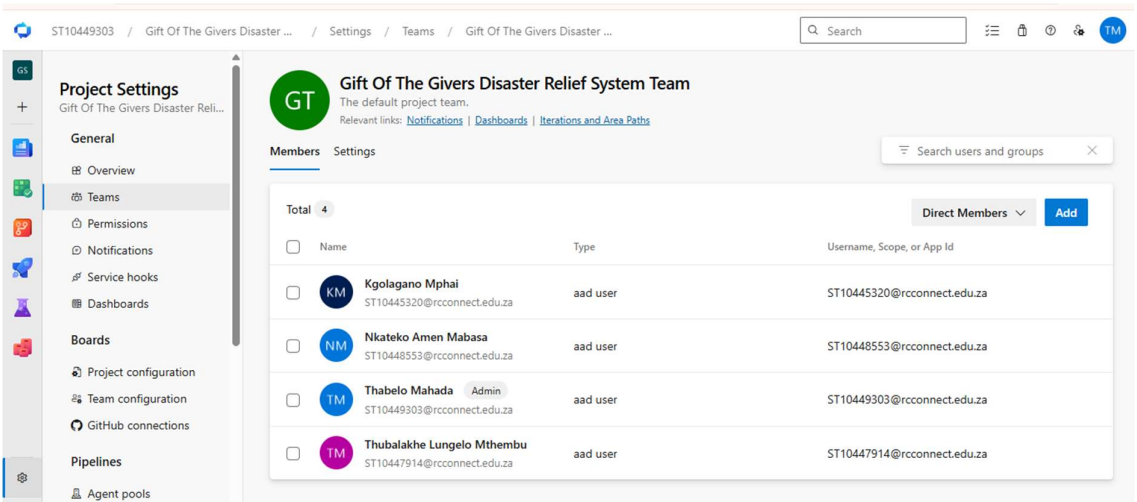
28 AUGUST 2026

Section A: Agile Project Planning with Azure Boards

1.1 Setting up Your Azure Boards Project

a. Add all group members and define their roles.

Figure 1: Azure DevOps team members for the Gift of the Givers Disaster Relief System.



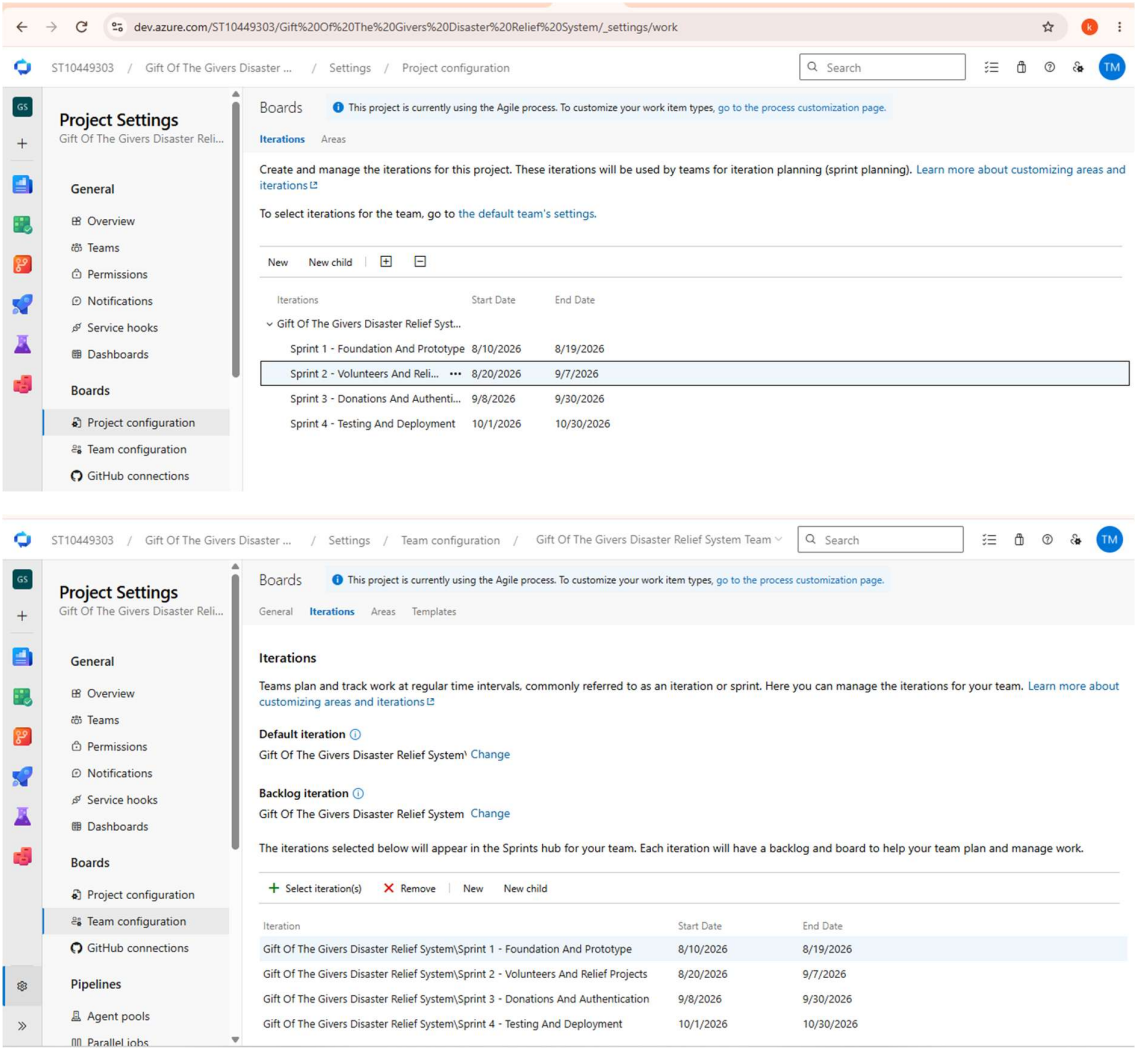
Member	Project Role	Responsibility
Kgolagano Mphai	Project Lead	Sprint planning, coordination and Azure Boards
Nkateko Mabasa	Database Designer	ERD and Azure SQL Database
Thabelo Mahada	Developer	ASP.NET Core development
Thubalakhe Mthembu	Quality Reviewer	Testing, bugs and documentation

Role	Azure DevOps Access
Project Lead	Project Administrator / appropriate project management access
Database Designer	Contributor
Developer	Contributor
Quality Reviewer	Contributor

b. Establish iteration paths (sprints) and area paths (modules of the application).

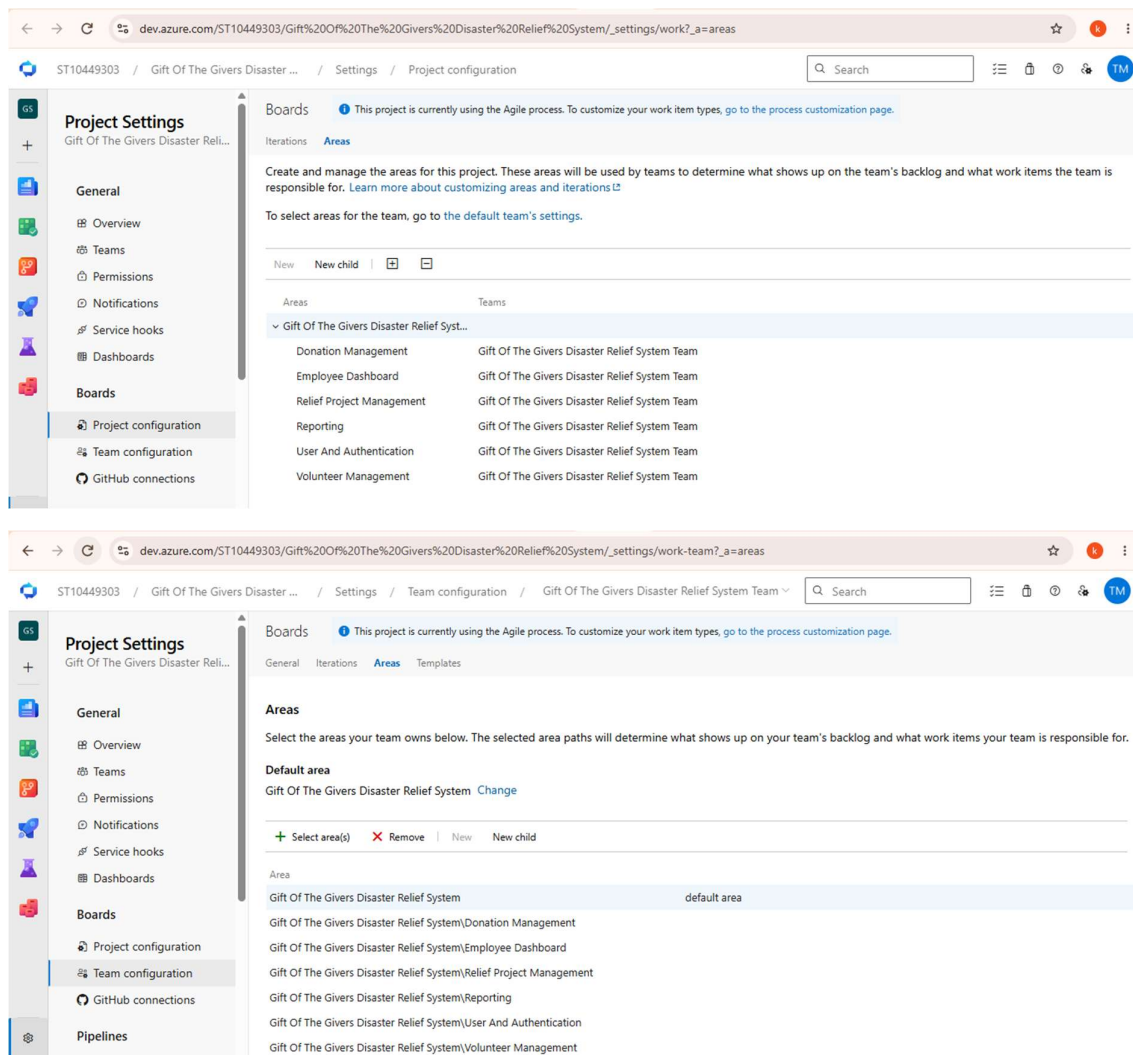
Iteration paths (sprints) screenshots

Figure 3: Azure DevOps iteration paths showing the planned development sprints and their dates.



Area paths (modules of the application) screenshots

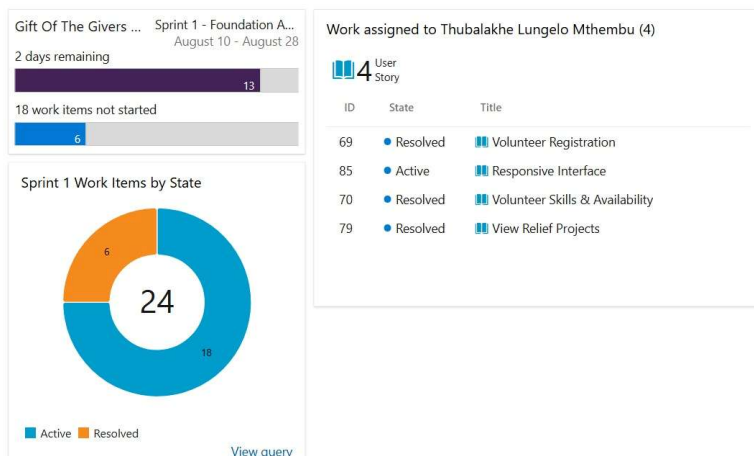
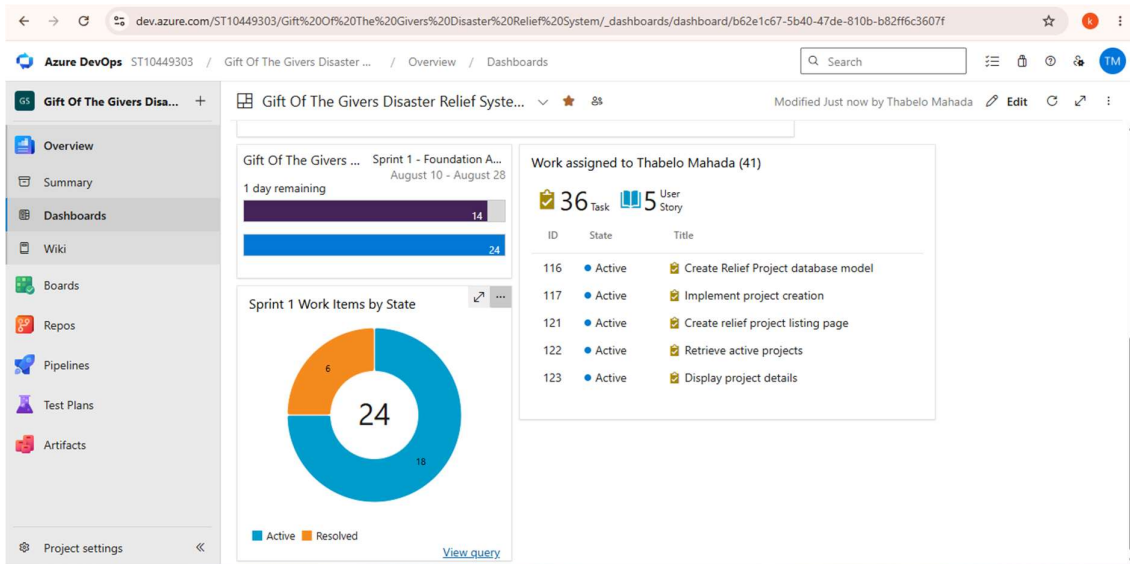
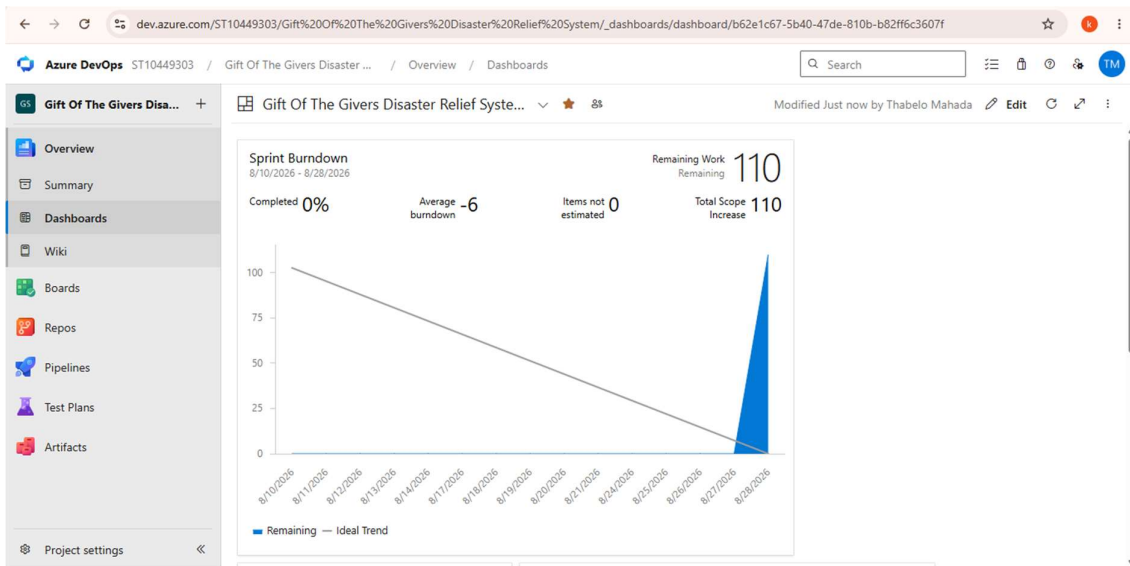
Figure 2: Azure DevOps Area Paths representing the major functional modules of the disaster relief application.

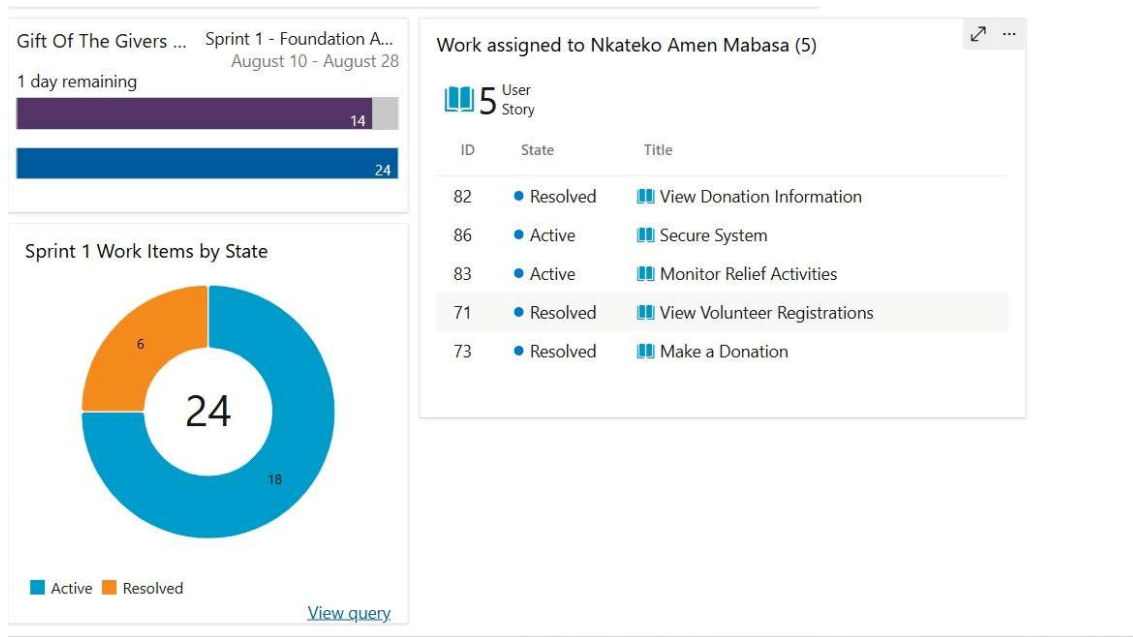


c. Provide screenshots and a short explanation of how the setup reflects real collaboration in agile teams.

Dashboard

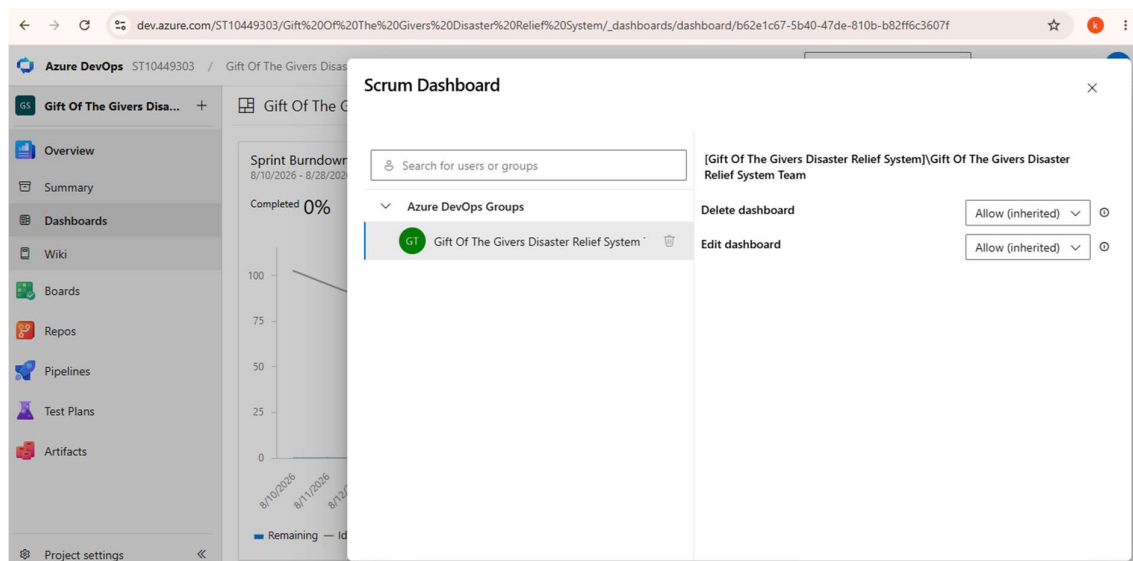
Figure 4: Agile project dashboard showing Sprint 1 progress, work-item status and remaining work. The dashboard provides a visual overview of the team's current sprint activities, including active and resolved work items and sprint burndown progress.





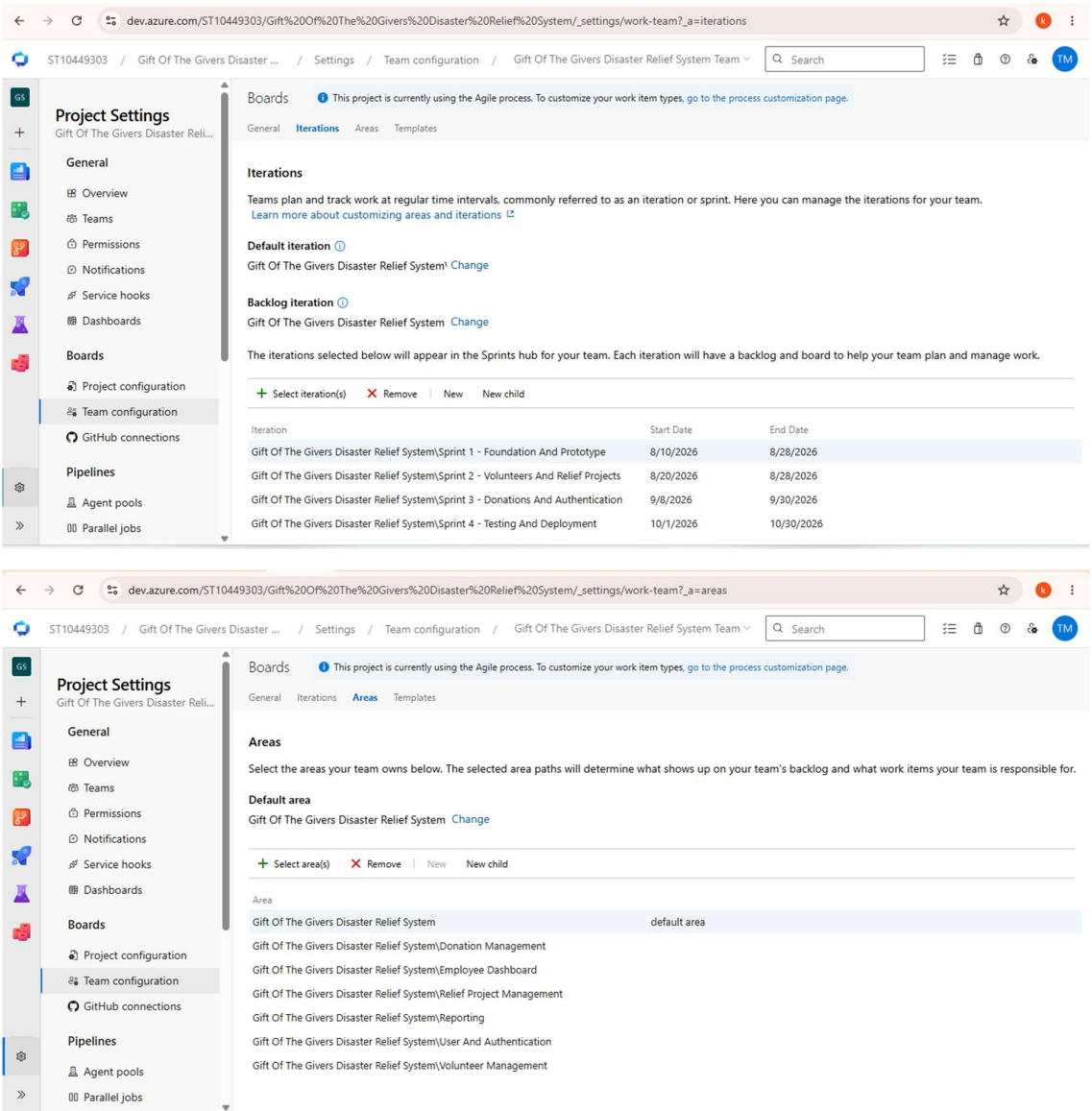
Dashboard Security/Permissions

Figure 5: Dashboard security permissions configured according to team responsibilities. Azure DevOps dashboard permissions control who can manage and edit the project dashboard, supporting appropriate access and collaboration within the Agile development team.



Team Configuration

Figure 6: Team configuration showing the Areas and Iterations available to the Agile development team. The configuration organises project work into defined areas and sprints, allowing work items to be assigned to the appropriate team and iteration.



d. Explanation of how the setup reflects real collaboration in agile teams.

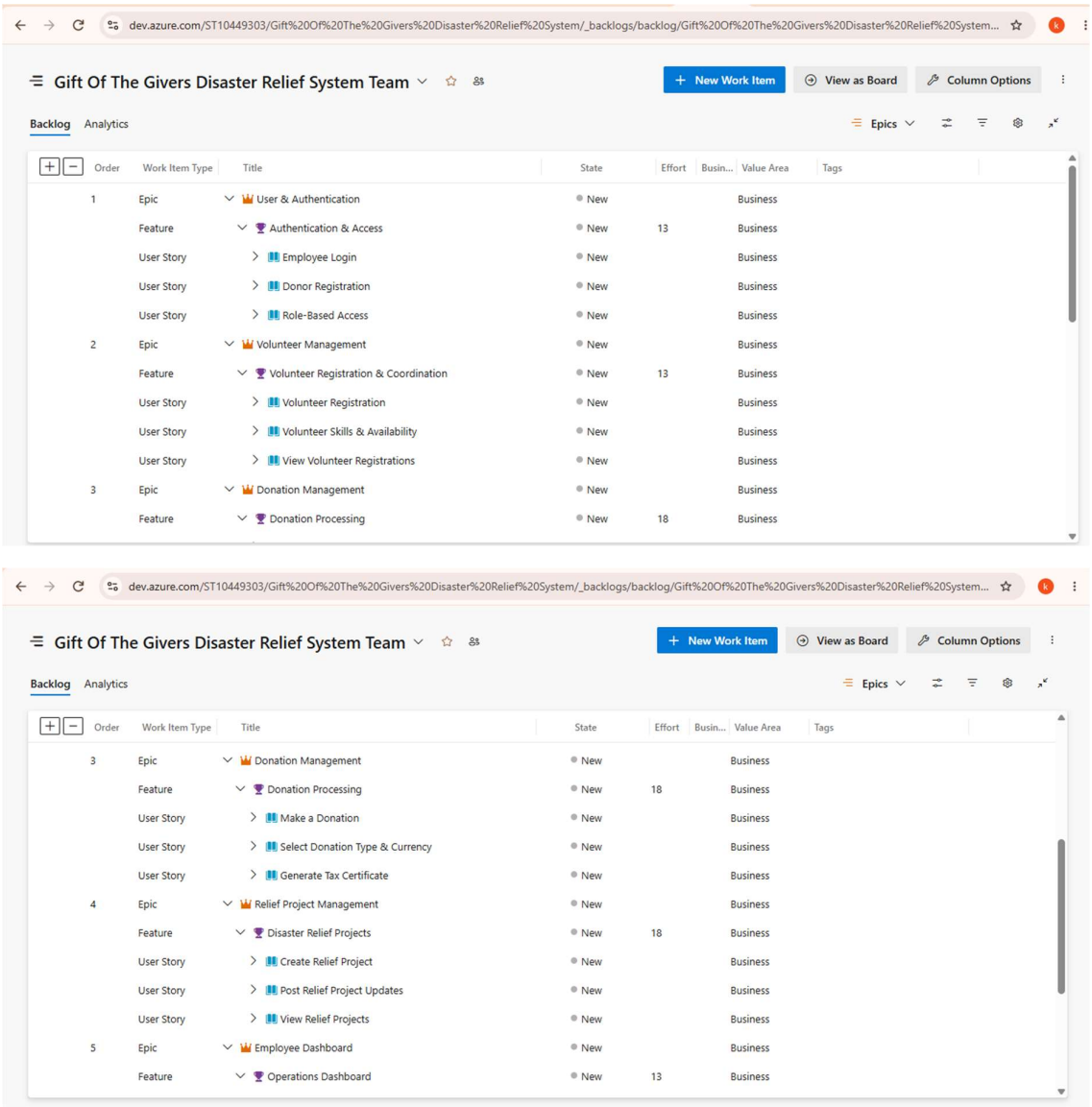
The Azure DevOps setup reflects real Agile collaboration by allowing team members to work together on shared tasks and monitor progress. Work items are assigned to specific team members and organised into Areas and Iterations, which helps the team understand who is responsible for each task, and which sprint it belongs to.

The Sprint Board provides a visual view of work-item status, such as New, Active, Resolved and Closed. The Sprint Burndown helps the team monitor remaining work and sprint progress. Dashboard permissions also ensure that team members have appropriate access while administrators can manage the project.

Overall, the setup supports transparency, accountability, communication and continuous progress, which are important principles of Agile teamwork.

1.2 Capturing Requirements as Epics, User Stories, and Tasks

Figure 7: Azure Boards backlog showing the hierarchy of Epics, Features and User Stories



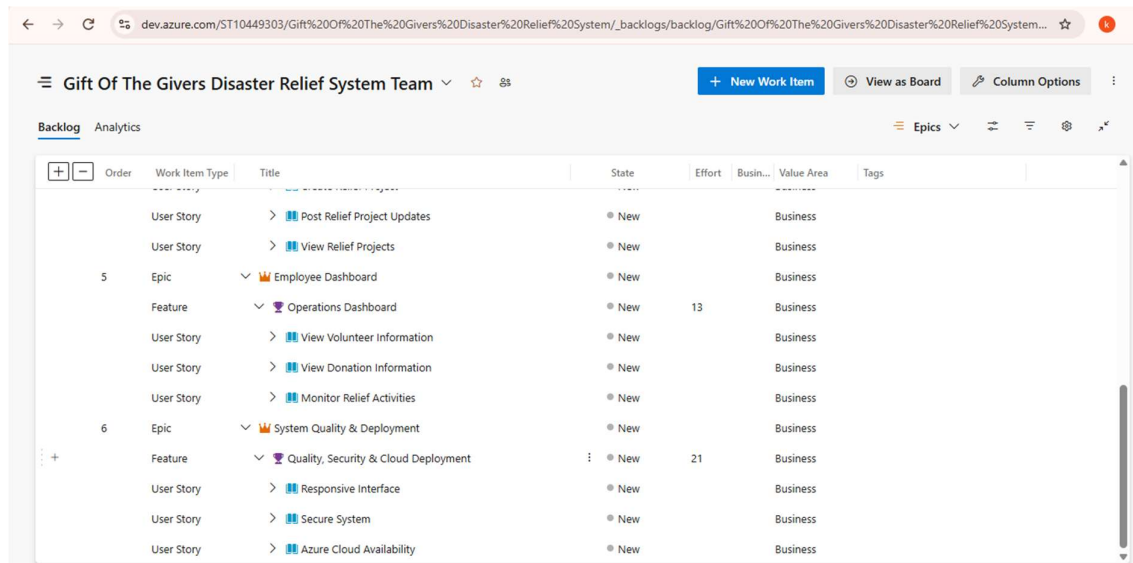
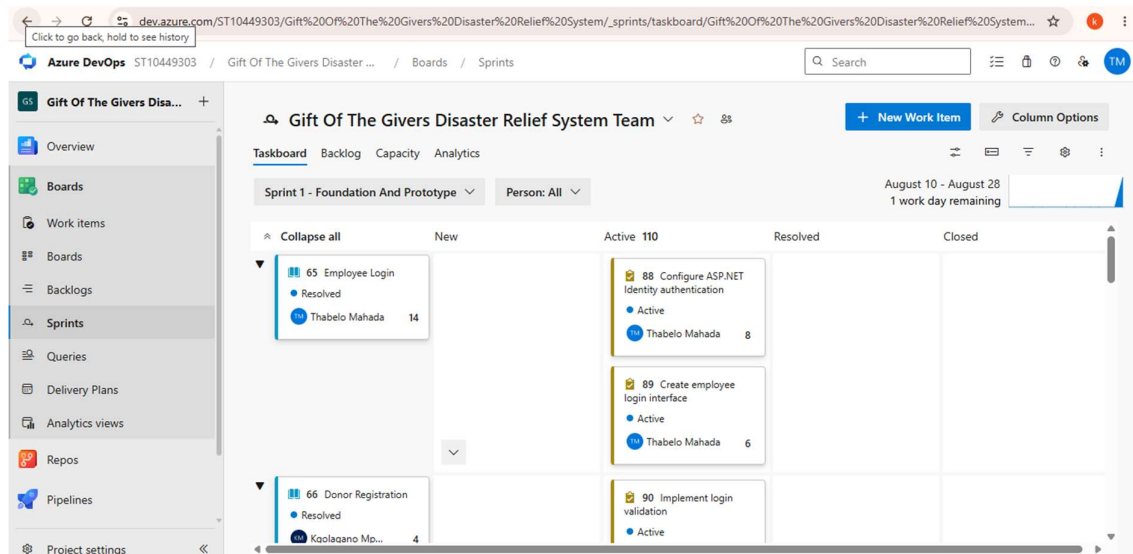


Figure 8: Azure Boards showing the status and assignment of project work items.



dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

Collapse all New Active 110 Resolved Closed

66 Donor Registration
Resolved
Kgolagano Mp... 4

73 Make a Donation
Resolved
Nkateko Amen... 38

90 Implement login validation
Active
Thabelo Mahada 4

91 Create donor registration form
Active
Thabelo Mahada 6

92 Configure donor account creation
Active
Thabelo Mahada

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

Collapse all New Active 110 Resolved Closed

93 Validate registration information
Active
Thabelo Mahada 8

106 Create donation form
Active
Thabelo Mahada 6

107 Create Donation database model
Active
Thabelo Mahada

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

Collapse all New Active 110 Resolved Closed

74 Select Donation Type & Currency
Resolved
Kgolagano Mp... 18

108 Implement donation submission
Active
Thabelo Mahada 4

109 Add one-time and recurring donation options
Active
Thabelo Mahada 6

110 Add ZAR, USD and EUR currency options
Active
Thabelo Mahada 8

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

Collapse all New Active 110 Resolved Closed

111 Validate donation selections
Active
Thabelo Mahada 4

77 Create Relief Project
Resolved
Thabelo Mahada 6

115 Create relief project form
Active
Thabelo Mahada 6

79 View Relief Projects
Resolved
Thubalakhe Lu... 30

116 Create Relief Project database model
Active
Thabelo Mahada 8

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

Collapse all New Active 110 Resolved Closed

117 Implement project creation
Active
Thabelo Mahada 4

121 Create relief project listing page
Active
Thabelo Mahada 8

122 Retrieve active projects
Active
Thabelo Mahada 6

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 28
1 work day remaining

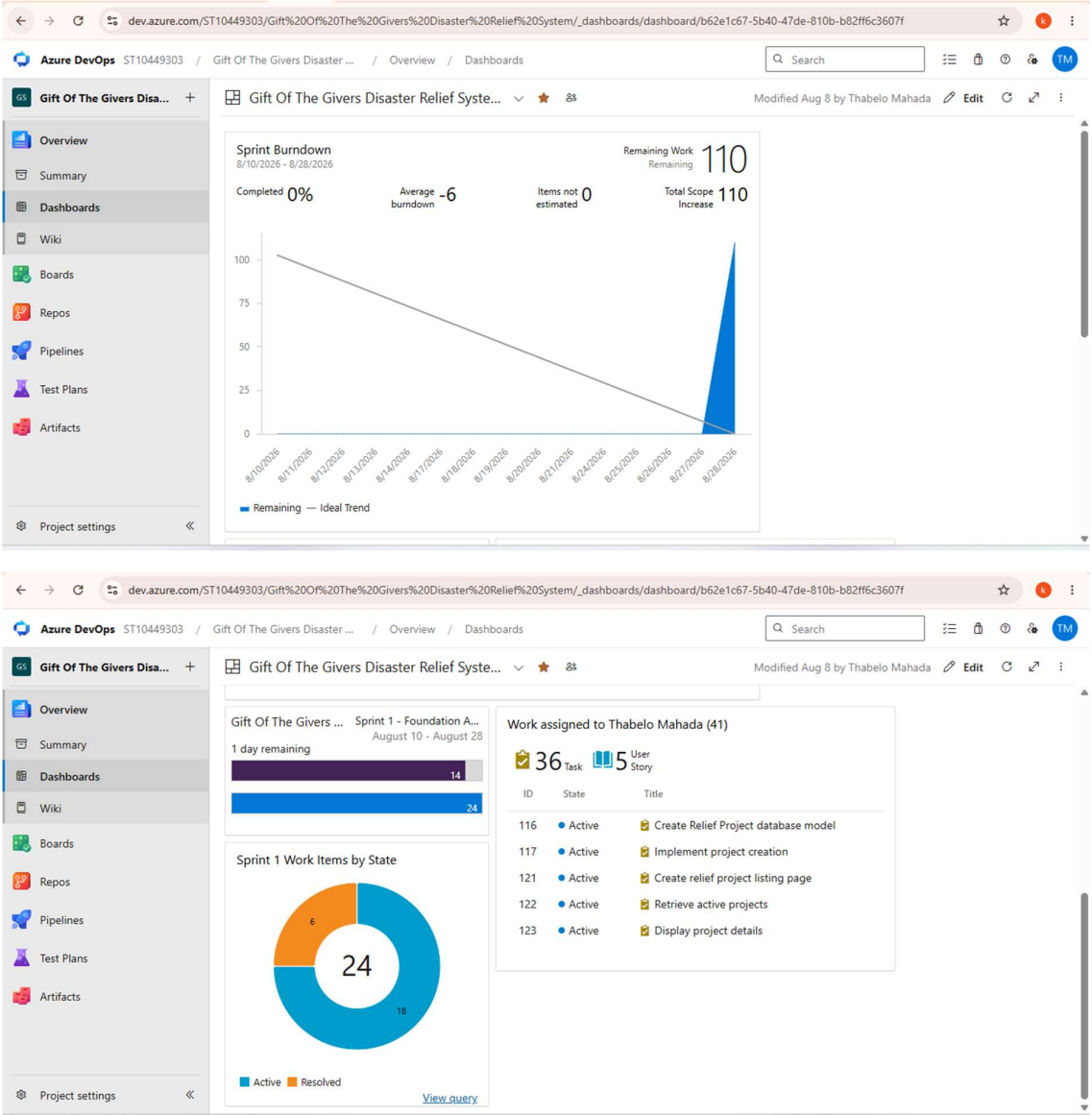
Collapse all New Active 110 Resolved Closed

listing page
Active
Thabelo Mahada 8

122 Retrieve active projects
Active
Thabelo Mahada 6

123 Display project details
Active
Thabelo Mahada 4

Figure 9: Azure DevOps dashboard showing sprint progress, assigned work and work-item status.



b. Why did we structure the work this way

Rationale for Work-Item Structure

The Gift of the Givers Disaster Relief System was divided into Epics based on the major functional and operational areas of the application. Features were used to group related functionality within each Epic, while User Stories describe specific requirements from the perspective of the system's users. Tasks represent the technical activities required to implement each User Story.

This hierarchical structure allows the team to break large and complex requirements into smaller, manageable pieces of work that can be assigned and completed during sprints (Microsoft Ignite, 2026).

Rationale for Prioritisation

Work items were prioritised according to their importance to the disaster relief application. Core functionality such as authentication, volunteer management, donations and relief project management was given a higher priority because these functions are essential to the operation of the system. Supporting functionality such as dashboard monitoring and responsive design was given a lower priority so that the team can focus on delivering critical functionality first (Microsoft Ignite, 2026).

Rationale for Using Backlogs, Boards and Dashboards

Azure Boards Backlogs are used to organise requirements and show the relationship between Epics, Features, User Stories and Tasks. Boards are used to track the progress of assigned work through different development states. Dashboards provide a high-level view of project progress and allow the team to monitor sprint performance and outstanding work. Together, these tools support transparency, collaboration and continuous progress within the Agile team (Microsoft Ignite, 2026).

1.3 Sprint Organisation

a. Define the Sprint Goals

Sprint 1 – Foundation And Prototype

Goal:

Develop a functional prototype demonstrating the main user journeys, including employee login, donor registration, volunteer registration, donations and basic relief project management.

Deliverables:

- Basic ASP.NET Core web application
- Gift of the Givers branding and navigation
- Employee login
- Donor registration
- Volunteer registration
- Donation functionality
- Donation type and currency selection

- Basic relief project creation
- Relief project viewing

Sprint 2 – Volunteers And Relief Projects

Goal:

Expand the prototype by implementing volunteer coordination and relief project management functionality.

Deliverables:

- Volunteer skills and availability
- Volunteer registration management
- Volunteer information
- Relief project updates
- Relief activity monitoring

Sprint 3 – Donations And Authentication

Goal:

Expand the system's donation management and authentication capabilities through role-based access, donation information and tax certificate functionality.

Deliverables:

- Role-based access
- Employee and donor permissions
- Donation records
- Tax certificate generation
- Donation information for employees

Sprint 4 – Testing And Deployment

Goal:

Finalise, test, secure and deploy the Gift of the Givers Disaster Relief System.

Deliverables:

- Responsive interface

- Security testing
- Application testing
- Bug fixing
- Authentication and authorisation testing
- Azure App Service deployment
- Live application URL

Screenshots

Screenshot A — Sprint 1

Figure X: Sprint 1 – Foundation and Prototype showing the User Stories selected for the initial working prototype

The figure consists of two screenshots of the Azure DevOps interface, specifically the 'Sprint 1 - Foundation And Prototype' taskboard. Both screenshots show the same project, 'Gift Of The Givers Disaster Relief System Team', and the same sprint, 'Sprint 1 - Foundation And Prototype', which runs from August 10 to August 19 with 8 work days remaining. The taskboard is organized into columns: New, Active, Resolved, and Closed. The left sidebar shows the project navigation menu with 'Sprints' selected. The top navigation bar includes 'New Work Item' and 'Column Options' buttons. The bottom screenshot shows a different set of user stories than the top one.

Screenshot	User Story ID	Title	Assignee	Status
Top	65	Employee Login	Thabelo Mahada	New
Top	88	Configure ASP.NET Identity authentication	Unassigned	New
Top	89	Create employee login interface	Unassigned	New
Top	90	Implement login validation	Unassigned	New
Bottom	66	Donor Registration	Kgolagano Mphai	New
Bottom	91	Create donor registration form	Unassigned	New
Bottom	92	Configure donor account creation	Unassigned	New
Bottom	93	Validate registration information	Unassigned	New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

+ New Work Item Column Options

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 19
8 work days remaining

Collapse all

New

Active

Resolved

Closed

69 Volunteer Registration
New
Thubalakhe Lungelo ...

97 Design volunteer registration form
New
Unassigned

98 Create Volunteer model/database table
New
Unassigned

99 Implement volunteer registration
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

+ New Work Item Column Options

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 19
8 work days remaining

Collapse all

New

Active

Resolved

Closed

73 Make a Donation
New
Nkateko Amen Mabasa

106 Create donation form
New
Unassigned

107 Create Donation database model
New
Unassigned

108 Implement donation submission
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

+ New Work Item Column Options

Taskboard Backlog Capacity Analytics

Sprint 1 - Foundation And Prototype Person: All

August 10 - August 19
8 work days remaining

Collapse all

New

Active

Resolved

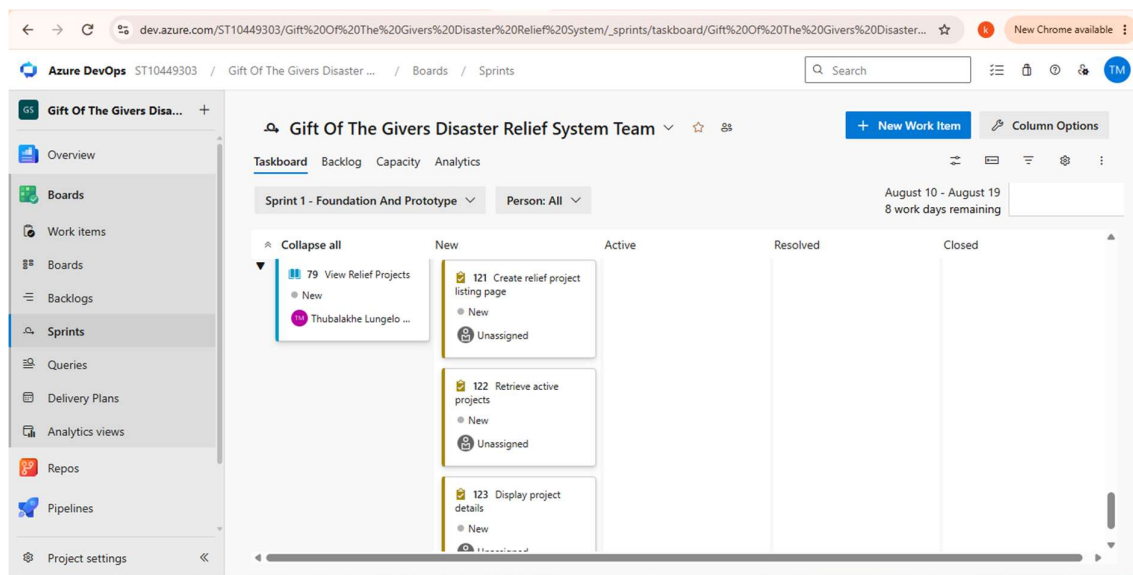
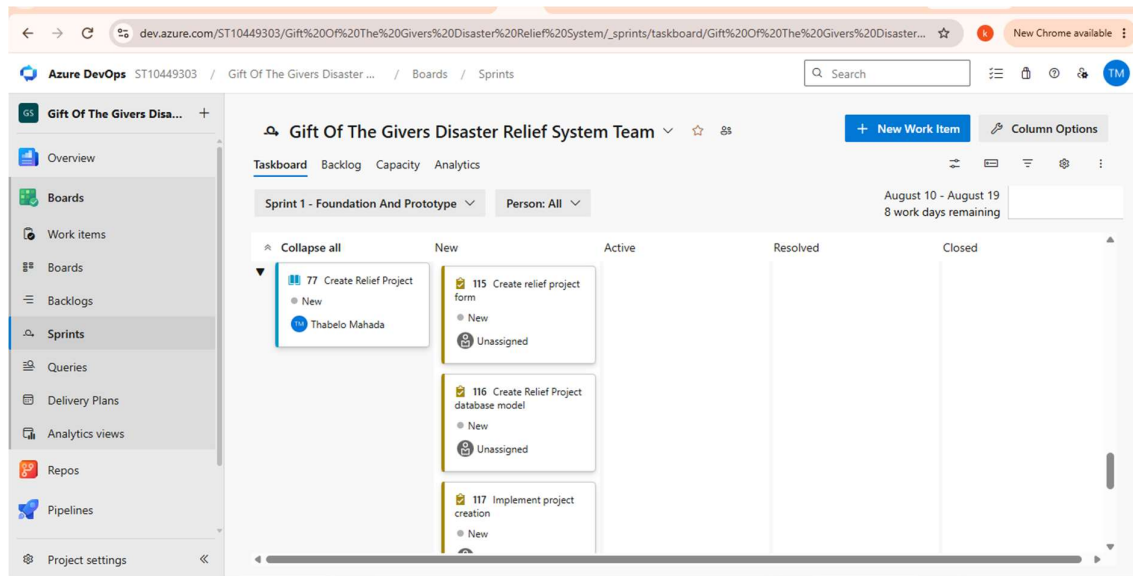
Closed

74 Select Donation Type & Currency
New
Kgolagano Mphai

109 Add one-time and recurring donation options
New
Unassigned

110 Add ZAR, USD and EUR currency options
New
Unassigned

111 Validate donation selections
New



Screenshot B — Sprint 2

Figure X: Sprint 2 – Volunteers and Relief Projects showing the expansion of volunteer coordination and relief project functionality.

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 2 - Volunteers And Relief Projects Person: All

August 20 - September 7 13 work days

Collapse all New Active Resolved Closed

70 Volunteer Skills & Availability
New
Thubalakhe Lungelo ...

100 Add skills and availability fields
New
Unassigned

101 Update volunteer database model
New
Unassigned

102 Implement skills and availability submission
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 2 - Volunteers And Relief Projects Person: All

August 20 - September 7 13 work days

Collapse all New Active Resolved Closed

71 View Volunteer Registrations
New
Nkateko Amen Mabasa

103 Create employee volunteer list
New
Unassigned

104 Retrieve volunteer records from database
New
Unassigned

105 Implement volunteer filtering/viewing
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 2 - Volunteers And Relief Projects Person: All

August 20 - September 7 13 work days

Collapse all

New Active Resolved Closed

78 Post Relief Project Updates
New
Thabelo Mahada

118 Create project update interface
New
Unassigned

119 Implement update submission
New
Unassigned

120 Display project updates
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 2 - Volunteers And Relief Projects Person: All

August 20 - September 7 13 work days

Collapse all

New Active Resolved Closed

81 View Volunteer Information
New
Kgolagano Mphai

124 Create volunteer dashboard section
New
Unassigned

125 Retrieve volunteer statistics
New
Unassigned

126 Display volunteer information
New

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disaster Relief System Team

Taskboard Backlog Capacity Analytics

Sprint 2 - Volunteers And Relief Projects Person: All

August 20 - September 7 13 work days

Collapse all

New Active Resolved Closed

83 Monitor Relief Activities
New
Nkateko Amen Mabasa

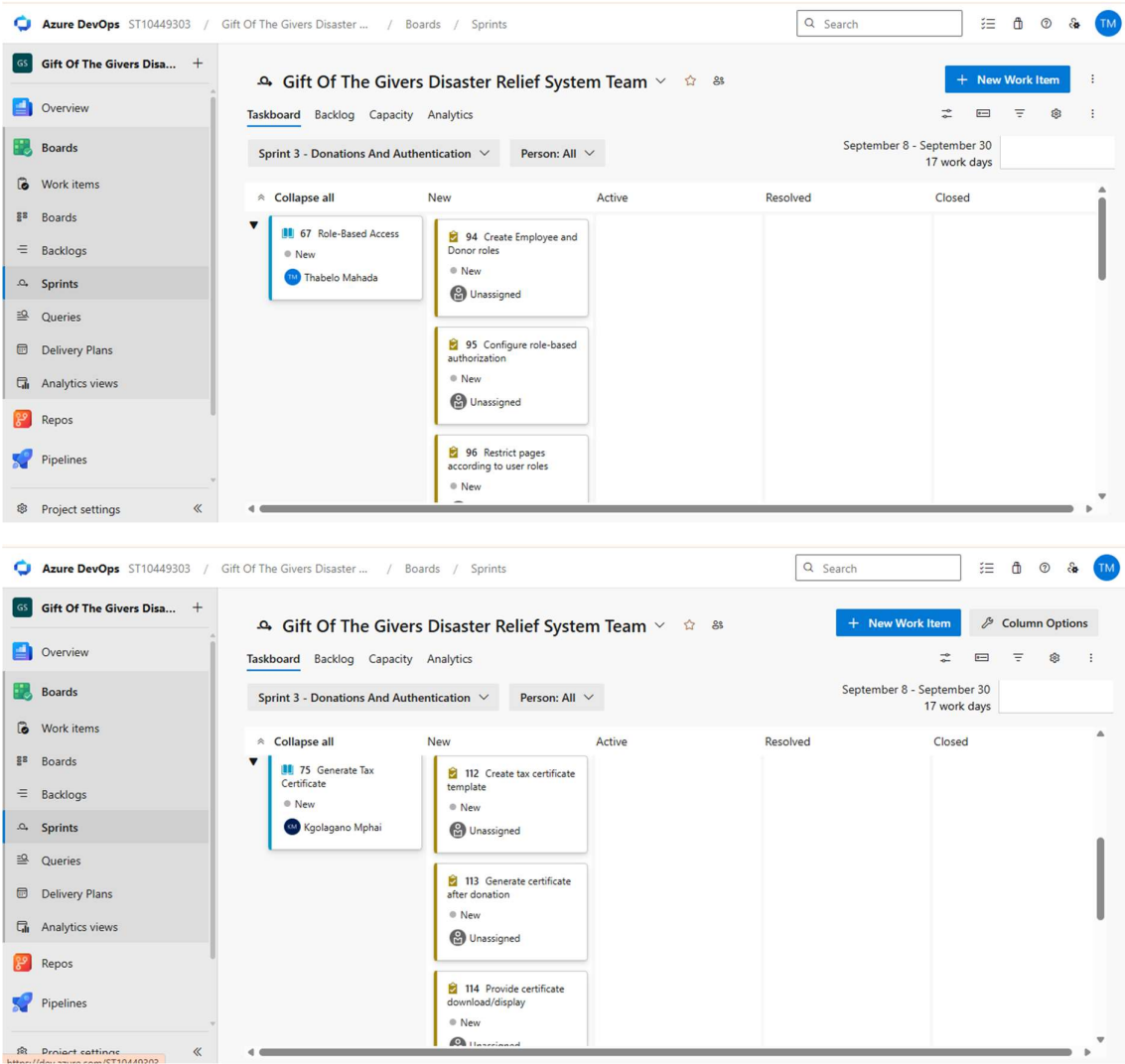
130 Create relief activity dashboard section
New
Unassigned

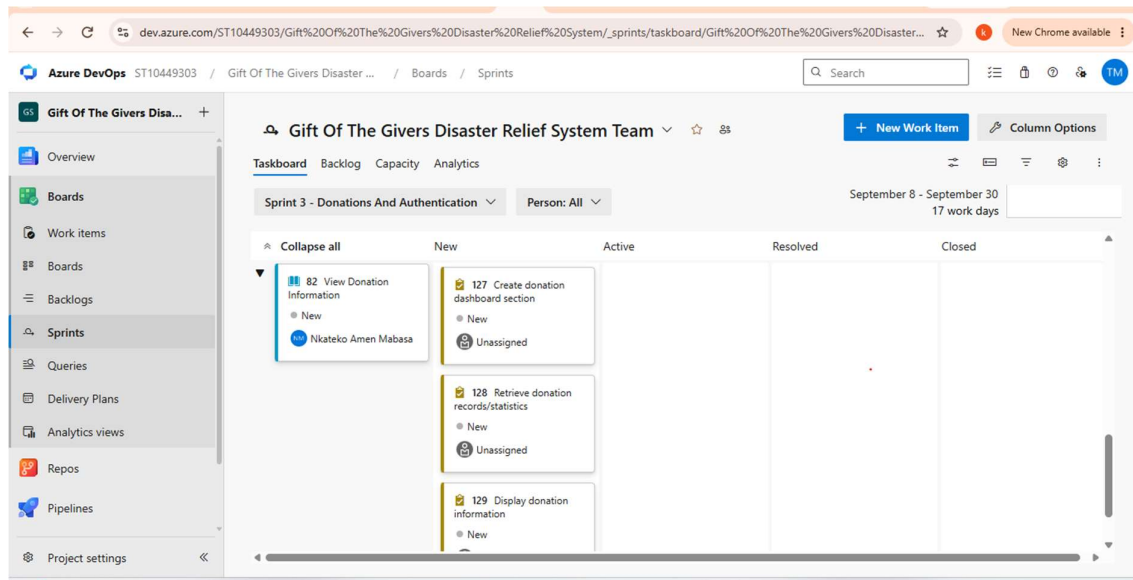
131 Retrieve project activity information
New
Unassigned

132 Display current relief activities
New

Screenshot C — Sprint 3

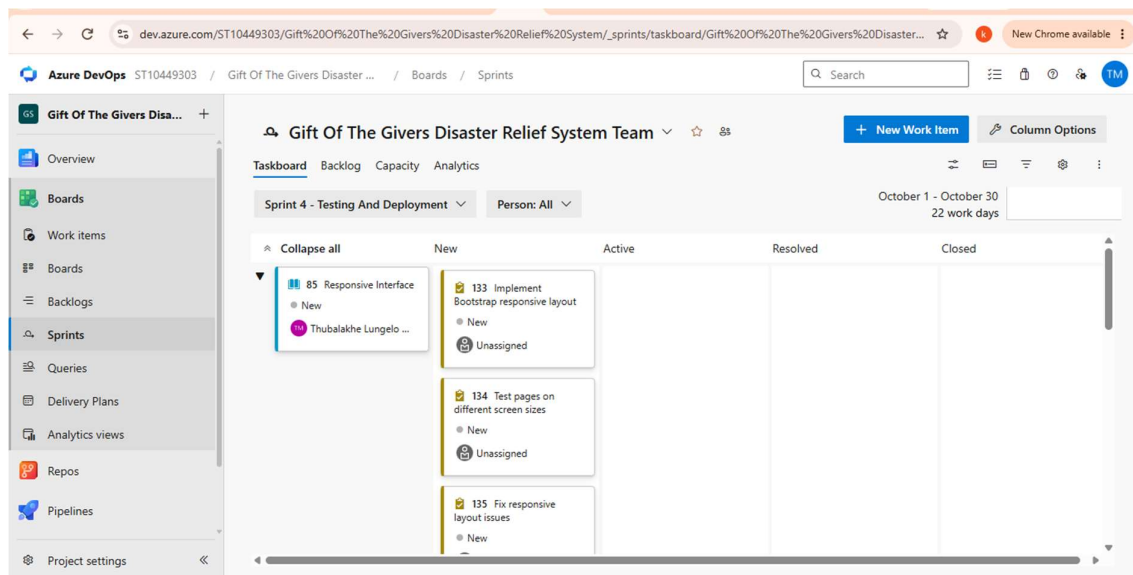
Figure X: Sprint 3 – Donations and Authentication showing the expansion of donation management and role-based access.





Screenshot D — Sprint 4

Figure X: Sprint 4 – Testing and Deployment showing final testing, security and Azure deployment activities.



dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disa... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Project settings

Gift Of The Givers Disaster Relief System Team

+ New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Sprint 4 - Testing And Deployment Person: All

October 1 - October 30
22 work days

Collapse all

New

Active

Resolved

Closed

86 Secure System
New
Nkateko Amen Mabasa

136 Configure authentication and authorisation security
New
Unassigned

137 Protect sensitive user and donation information
New
Unassigned

138 Test unauthorized access

dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/_sprints/taskboard/Gift%20Of%20The%20Givers%20Disaster...

Azure DevOps ST10449303 / Gift Of The Givers Disaster ... / Boards / Sprints

Gift Of The Givers Disa... +

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Pipelines

Project settings

Gift Of The Givers Disaster Relief System Team

+ New Work Item

Column Options

Taskboard Backlog Capacity Analytics

Sprint 4 - Testing And Deployment Person: All

October 1 - October 30
22 work days

Collapse all

New

Active

Resolved

Closed

87 Azure Cloud Availability
New
Thabelo Mahada

139 Configure Azure App Service
New
Unassigned

140 Deploy ASP.NET Core application
New
Unassigned

141 Test deployed application
New

Section B: Designing the Azure SQL Database

2.1 Why Database Design Matters

Database Design Principles

Good database design is critical for the Gift of the Givers Disaster Relief System because the application must manage large amounts of volunteer, donor, donation and relief-project data accurately and efficiently during emergencies.

Scalability: The database should be able to handle thousands of volunteers registering during a disaster without becoming slow or unreliable. Azure SQL can support increasing amounts of data, while properly designed tables and indexes allow frequently accessed information, such as volunteer registrations and relief projects, to be retrieved efficiently (GeeksForGeeks, 2026).

Data integrity: Primary keys (PKs) uniquely identify records, such as VolunteerID and DonationID, preventing duplicate identification. Foreign keys (FKs) connect related entities, such as linking a Donation to a Donor or a Volunteer to a ReliefProject. Referential integrity prevents invalid relationships and orphan records (GeeksForGeeks, 2026).

Efficiency: Normalisation reduces unnecessary duplication across tables, making updates more reliable. Indexes on frequently searched fields, such as donation dates, volunteer availability and project status, can improve query performance. This allows employees to quickly identify available volunteers, monitor donations and locate active relief projects when rapid decisions are required (GeeksForGeeks, 2026).

2.2 Creating the Database Schema

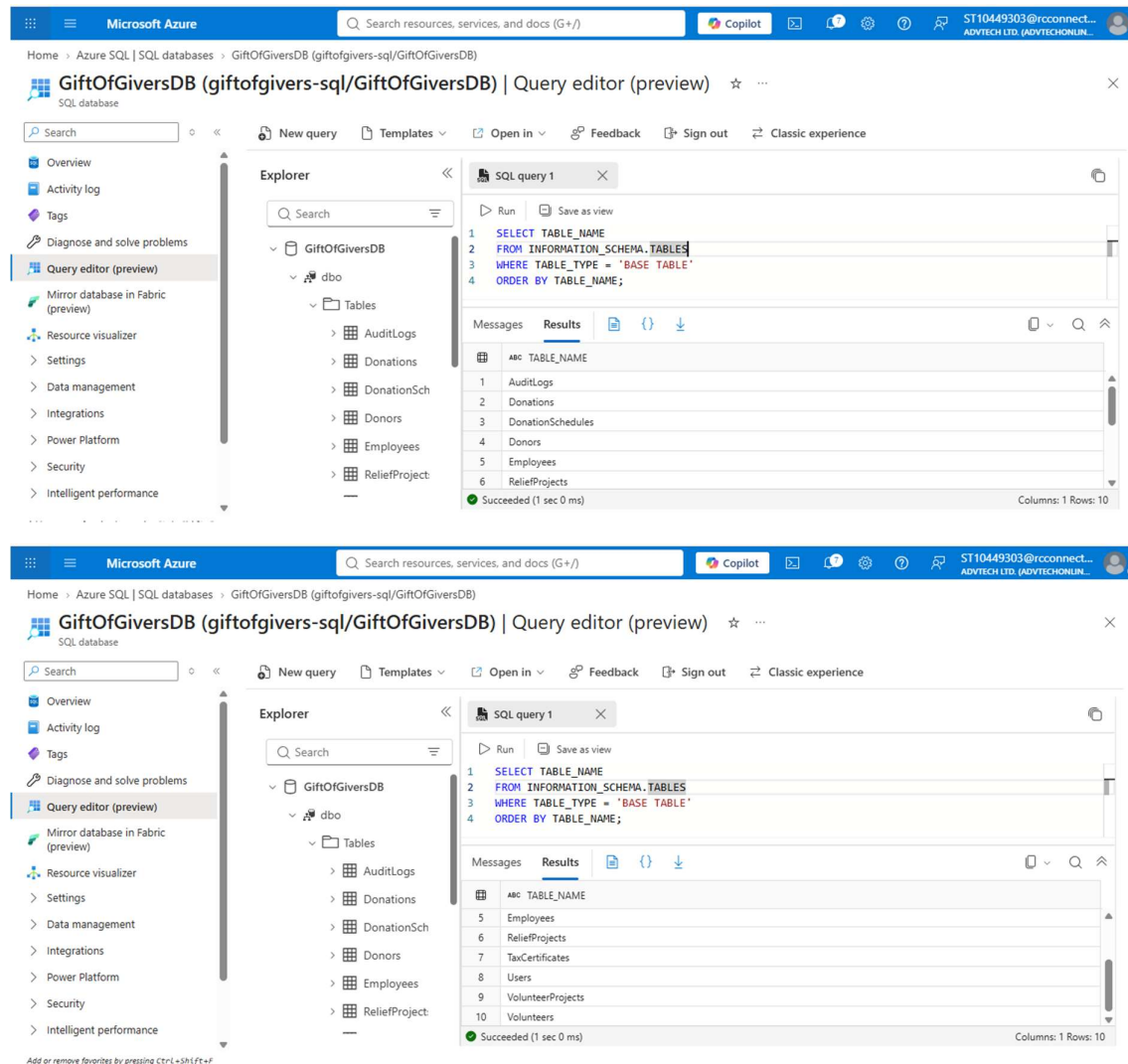
1. Entities

1. Users
2. Donors
3. Employees
4. Volunteers
5. ReliefProjects
6. Donations
7. DonationSchedules
8. TaxCertificates
9. VolunteerProjects

10.AuditLogs

Screenshot of your tables

Figure X: Tables created in the GiftOfGiversDB Azure SQL Database.



2. Attributes for each entity (with appropriate data types).

Screenshot showing attributes and data types

Figure X: Database entities, attributes and data types for the Gift of the Givers application.

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

DonationSch
Donors
Employees
ReliefProject
TaxCertificate
Users
VolunteerPrc
Volunteers
Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
1	Users		UserID		int		NULL		NO
2	Users		Email		nvarchar		255		NO
3	Users		PasswordHash		nvarchar		255		NO
4	Users		Role		nvarchar		50		NO
5	Users		IsActive		bit		NULL		NO
6	Users		CreatedAt		datetime2		NULL		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

DonationSch
Donors
Employees
ReliefProject
TaxCertificate
Users
VolunteerPrc
Volunteers
Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
6	Users		CreatedAt		datetime2		NULL		NO
7	Donors		DonorID		int		NULL		NO
8	Donors		UserID		int		NULL		NO
9	Donors		FirstName		nvarchar		100		NO
10	Donors		LastName		nvarchar		100		NO
11	Donors		PhoneNumber		nvarchar		20		YES

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

DonationSch
Donors
Employees
ReliefProject
TaxCertificate
Users
VolunteerPrc
Volunteers
Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
11	Donors		PhoneNumber		nvarchar		20		YES
12	Donors		CreatedAt		datetime2		NULL		NO
13	Employees		EmployeeID		int		NULL		NO
14	Employees		UserID		int		NULL		NO
15	Employees		FirstName		nvarchar		100		NO
16	Employees		LastName		nvarchar		100		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

ST10449303@rconnect... ADVTECH LTD. (ADVTECHONLINE...)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance

Explorer

Search

DonationSch Donors Employees ReliefProject TaxCertificate Users VolunteerPrc Volunteers Views

SQL query 1

```
29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;
```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
16	Employees		LastName		nvarchar	100			NO
17	Employees		JobTitle		nvarchar	100			YES
18	Employees		CreatedAt		datetime2				NO
19	Volunteers		VolunteerID		int				NO
20	Volunteers		UserID		int				YES
21	Volunteers		FirstName		nvarchar	100			NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

ST10449303@rconnect... ADVTECH LTD. (ADVTECHONLINE...)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance

Explorer

Search

DonationSch Donors Employees ReliefProject TaxCertificate Users VolunteerPrc Volunteers Views

SQL query 1

```
29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;
```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
21	Volunteers		FirstName		nvarchar	100			NO
22	Volunteers		LastName		nvarchar	100			NO
23	Volunteers		Email		nvarchar	255			NO
24	Volunteers		PhoneNumber		nvarchar	20			YES
25	Volunteers		Skills		nvarchar	500			YES
26	Volunteers		Availability		nvarchar	255			YES

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

ST10449303@rconnect... ADVTECH LTD. (ADVTECHONLINE...)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Resource visualizer Settings Data management Integrations Power Platform Security Intelligent performance

Explorer

Search

DonationSch Donors Employees ReliefProject TaxCertificate Users VolunteerPrc Volunteers Views

SQL query 1

```
29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;
```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
26	Volunteers		Availability		nvarchar	255			YES
27	Volunteers		RegistrationDate		datetime2				NO
28	ReliefProjects		ProjectID		int				NO
29	ReliefProjects		EmployeeID		int				NO
30	ReliefProjects		ProjectName		nvarchar	200			NO
31	ReliefProjects		Description		nvarchar	1000			YES

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
31	ReliefProjects		Description		nvarchar		1000		YES
32	ReliefProjects		Location		nvarchar		200		NO
33	ReliefProjects		Status		nvarchar		50		NO
34	ReliefProjects		StartDate		date		NULL		YES
35	ReliefProjects		EndDate		date		NULL		YES
36	ReliefProjects		CreatedAt		datetime2		NULL		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
36	ReliefProjects		CreatedAt		datetime2		NULL		NO
37	Donations		DonationID		int		NULL		NO
38	Donations		DonorID		int		NULL		YES
39	Donations		ProjectID		int		NULL		YES
40	Donations		Amount		decimal		NULL		NO
41	Donations		Currency		nvarchar		3		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
41	Donations		Currency		nvarchar		3		NO
42	Donations		DonationType		nvarchar		20		NO
43	Donations		DonationDate		datetime2		NULL		NO
44	Donations		IsAnonymous		bit		NULL		NO
45	DonationSchedules		ScheduleID		int		NULL		NO
46	DonationSchedules		DonorID		int		NULL		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
46	DonationSchedules		DonorID		int		NULL		NO
47	DonationSchedules		ProjectID		int		NULL		YES
48	DonationSchedules		Amount		decimal		NULL		NO
49	DonationSchedules		Currency		nvarchar		3		NO
50	DonationSchedules		Frequency		nvarchar		20		NO
51	DonationSchedules		StartDate		date		NULL		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
51	DonationSchedules		StartDate		date		NULL		NO
52	DonationSchedules		EndDate		date		NULL		YES
53	DonationSchedules		IsActive		bit		NULL		NO
54	TaxCertificates		TaxCertificateID		int		NULL		NO
55	TaxCertificates		DonationID		int		NULL		NO
56	TaxCertificates		CertificateNumber		nvarchar		50		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance

Explorer

Search

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
56	TaxCertificates		CertificateNumber		nvarchar		50		NO
57	TaxCertificates		IssueDate		datetime2		NULL		NO
58	TaxCertificates		CertificateStatus		nvarchar		30		NO
59	VolunteerProjects		VolunteerID		int		NULL		NO
60	VolunteerProjects		ProjectID		int		NULL		NO
61	VolunteerProjects		AssignmentDate		datetime2		NULL		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
61	VolunteerProjects		AssignmentDate		datetime2		NULL		NO
62	VolunteerProjects		AssignmentStatus		nvarchar		30		NO
63	AuditLogs		AuditLogID		int		NULL		NO
64	AuditLogs		UserID		int		NULL		YES
65	AuditLogs		Action		nvarchar		100		NO
66	AuditLogs		TableName		nvarchar		100		NO

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

29 WHEN 'TaxCertificates' THEN 8
30 WHEN 'VolunteerProjects' THEN 9
31 WHEN 'AuditLogs' THEN 10
32 END,
33 ORDINAL_POSITION;

```

Messages Results

ABC	TABLE_NAME	ABC	COLUMN_NAME	ABC	DATA_TYPE	123	CHARACTER_MA...	ABC	IS_NULLABLE
64	AuditLogs		UserID		int		NULL		YES
65	AuditLogs		Action		nvarchar		100		NO
66	AuditLogs		TableName		nvarchar		100		NO
67	AuditLogs		RecordID		int		NULL		YES
68	AuditLogs		ActionDate		datetime2		NULL		NO
69	AuditLogs		Details		nvarchar		1000		YES

Succeeded (0 sec 667 ms) Columns: 5 Rows: 69

3. Primary and foreign key relationships.

Screenshot of Primary and Foreign Keys

Figure X: Foreign-key relationships between the database entities.

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB)

GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview)

SQL database

Search

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explorer

- DonationSch
- Donors
- Employees
- ReliefProject
- TaxCertificate
- Users
- VolunteerPrc
- Volunteers
- Views

SQL query 1

```

19 WHEN 'TaxCertificates' THEN 8
20 WHEN 'VolunteerProjects' THEN 9
21 WHEN 'AuditLogs' THEN 10
22 END,
23 fk.name;

```

Messages Results

ABC	ForeignKeyName	ABC	ChildTable	ABC	ForeignKeyColumn	ABC	ParentTable	ABC	PrimaryKeyColumn
1	FK_Donors_Users		Donors		UserID		Users		UserID
2	FK_Employees_Users		Employees		UserID		Users		UserID
3	FK_Volunteers_Users		Volunteers		UserID		Users		UserID
4	FK_ReliefProjects_Emp...		ReliefProjects		EmployeeID		Employees		EmployeeID
5	FK_Donations_Donors		Donations		DonorID		Donors		DonorID
6	FK_Donations_ReliefPr...		Donations		ProjectID		ReliefProjects		ProjectID

Succeeded (1 sec 476 ms) Columns: 5 Rows: 12

Microsoft Azure | Search resources, services, and docs (G+ /) | Copilot | ST10449303@rconnect... ADVTECH LTD. (ADVTECHONLIN...)

Home > Azure SQL | SQL databases > GiftOfGiversDB (giftofgivers-sql/GiftOfGiversDB) | Query editor (preview) ☆ ...

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Explorer

SQL query 1

```

19 WHEN 'TaxCertificates' THEN 8
20 WHEN 'VolunteerProjects' THEN 9
21 WHEN 'AuditLogs' THEN 10
22 END,
23 fk.name;

```

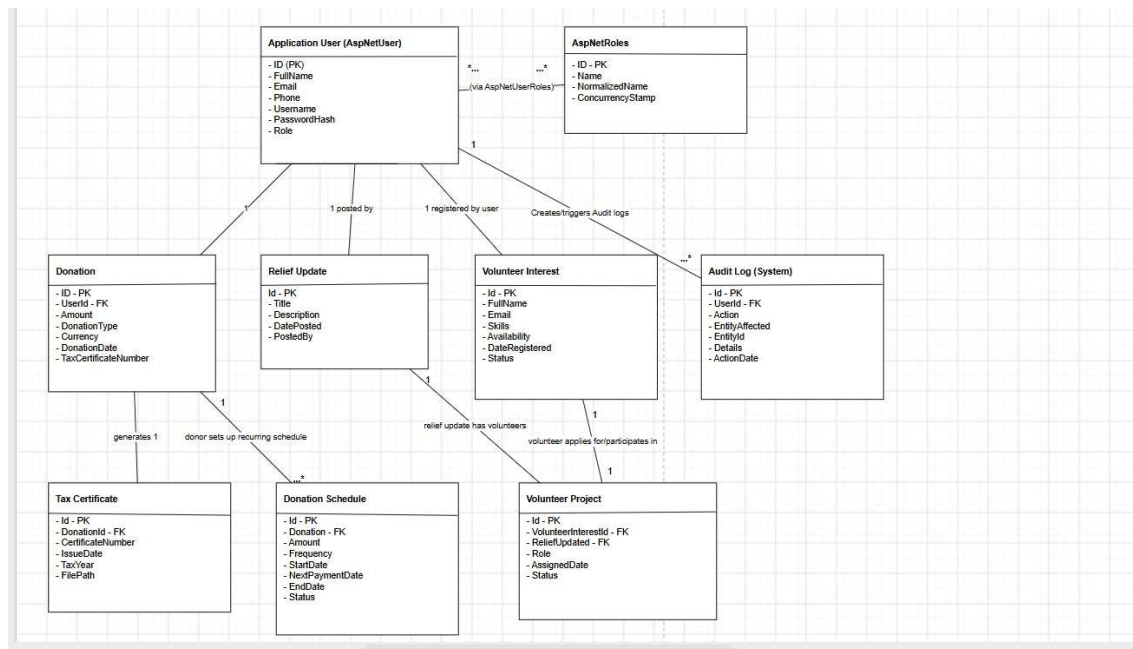
Messages Results

ABC	ForeignKeyName	ABC	ChildTable	ABC	ForeignKeyColumn	ABC	ParentTable	ABC	PrimaryKeyColumn
7	FK_DonationSchedule...		DonationSchedules		DonorID		Donors		DonorID
8	FK_DonationSchedule...		DonationSchedules		ProjectID		ReliefProjects		ProjectID
9	FK_TaxCertificates_Do...		TaxCertificates		DonationID		Donations		DonationID
10	FK_VolunteerProjects...		VolunteerProjects		ProjectID		ReliefProjects		ProjectID
11	FK_VolunteerProjects...		VolunteerProjects		VolunteerID		Volunteers		VolunteerID
12	FK_AuditLogs_Users		AuditLogs		UserID		Users		UserID

Succeeded (1 sec 476 ms) Columns: 5 Rows: 12

4. An ERD diagram showing how everything connects

Figure X: Entity Relationship Diagram (ERD) for the Gift of the Givers Disaster Relief System.



5. Provide explanations for each design choice: Why these entities? Why these relationships? How will this help the Foundation in practice?

Users

The Users entity stores common authentication and account information. It provides a central identity for people interacting with the system and supports role-based access such as employees, donors and volunteers.

Donors

The Donors entity stores donor-specific information separately from authentication details. This allows the Foundation to maintain donor information without duplicating user account data.

Employees

Employees stores information about Foundation staff who manage relief projects and monitor volunteer activity. Linking employees to Users supports role-based access.

Volunteers

Volunteers stores volunteer registration information such as skills and availability. This allows the Foundation to identify suitable volunteers quickly during disaster-response activities.

ReliefProjects

ReliefProjects represents individual disaster-relief initiatives. Each project can have responsible employees, volunteers and associated donations, allowing the Foundation to coordinate resources around specific relief efforts.

Donations

Donations records financial contributions, including amount, currency, donation type and whether the donation was anonymous. This is essential for tracking financial support received during relief operations.

DonationSchedules

DonationSchedules separates recurring donation information from individual donation transactions. This supports weekly, monthly or yearly donations while keeping the database normalised.

TaxCertificates

TaxCertificates stores the certificate information associated with donations. This supports the requirement for generating a placeholder tax certificate after a donation.

VolunteerProjects

VolunteerProjects is a junction table that connects volunteers with relief projects. It allows one volunteer to participate in multiple projects and one project to have many volunteers.

AuditLogs

AuditLogs records important system activities. This improves accountability and allows the Foundation to track changes made to important records.

Explain the relationships

The relationships were designed to maintain referential integrity and reduce duplication. Users are connected to donors, employees and volunteers so that authentication information is stored centrally. Employees are linked to relief projects to identify project responsibility. Donations are linked to donors and relief projects, allowing the Foundation to determine who contributed and which relief initiative received the funds. Recurring donations are stored in DonationSchedules, while TaxCertificates are linked directly to donations. The VolunteerProjects junction table resolves the many-to-many relationship between volunteers and relief projects. AuditLogs are linked to users to provide accountability for system activities.

2.3 Optimising Performance

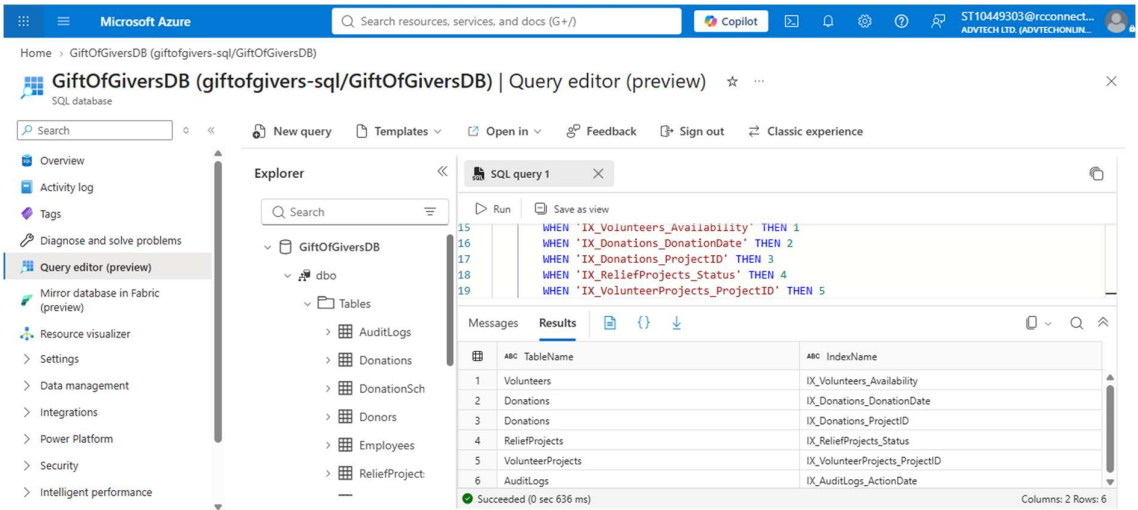
1. Indexing

Indexes were added to columns that are likely to be searched frequently. The Azure SQL Database can be optimised by adding **indexes** to frequently searched columns. In our database, indexes were added to volunteer availability, donation dates, donation project IDs, relief project status, volunteer project IDs and audit log dates. This helps the Foundation retrieve important information faster during disaster situations (Microsoft Learn, 2026).

We created:

```
CREATE INDEX IX_Volunteers_Availability
ON Volunteers(Availability);
CREATE INDEX IX_Donations_DonationDate
ON Donations(DonationDate);
CREATE INDEX IX_Donations_ProjectID
ON Donations(ProjectID);
CREATE INDEX IX_ReliefProjects_Status
ON ReliefProjects(Status);
CREATE INDEX IX_VolunteerProjects_ProjectID
ON VolunteerProjects(ProjectID);
CREATE INDEX IX_AuditLogs_ActionDate
ON AuditLogs(ActionDate);
```

Figure X: Performance indexes implemented in the GiftOfGiversDB Azure SQL Database



2. Which queries would benefit from optimisation?

The following queries would benefit from optimisation because they are likely to be used frequently during disaster-relief operations:

Finding available volunteers based on skills or availability.

Finding active relief projects by project status or location.

Retrieving donations for a specific relief project.

Retrieving recent donations using the donation date.

Finding volunteers assigned to a specific relief project.

These queries can be made faster by using the indexes created on Availability, Status, ProjectID, and DonationDate, and by selecting only the required columns instead of using SELECT *.

3. Would partitioning be useful for donations or volunteer records?

It can be useful for donations, potentially in the future. The Donations table could grow to millions of records. It could eventually be partitioned by **DonationDate**, allowing the database to manage large amounts of historical donation data more efficiently.

Section C: Azure App Services – Building a Prototype

3.1 .NET Framework vs .NET Core

3.1.1 Difference between .NET Framework and .NET Core

.NET Framework is Microsoft's older development platform designed mainly for Windows applications. It is closely tied to the Windows operating system and is commonly used for older web applications, desktop applications and legacy systems (GeeksForGeeks, 2023).

.NET Core was introduced as a modern, cross-platform alternative. It can run on **Windows, Linux and macOS**, and it is designed for modern web applications, APIs, containers and cloud environments such as Azure (GeeksForGeeks, 2023)

3.1.2 Which one is more suitable for a modern Azure-hosted application, and why?

For the Gift of the Givers application, .NET Core is more suitable for a modern Azure-hosted application.

Factor	.NET Framework	.NET Core
Portability	Mainly Windows	Cross-platform
Performance	Good	Generally better for modern web workloads
Cloud readiness	Less suited to modern cloud development	Designed with cloud/container workloads in mind
Azure hosting	Supported	Excellent fit
Modern development	Legacy-focused	Modern and actively developed

Practical reasoning

The Gift of the Givers application will to be hosted on **Azure App Services** and should be able to scale as the number of volunteers, donors and users increases. .NET Core is a better choice because it is lightweight, cross-platform and designed for modern web and cloud applications. Its portability also means the application is not restricted to a Windows-only environment.

3.2 Developing a Professional Prototype

3.2.1 Evidence: Prototype Pages

Figure 1: Gift of the Givers Disaster Relief System Home Page

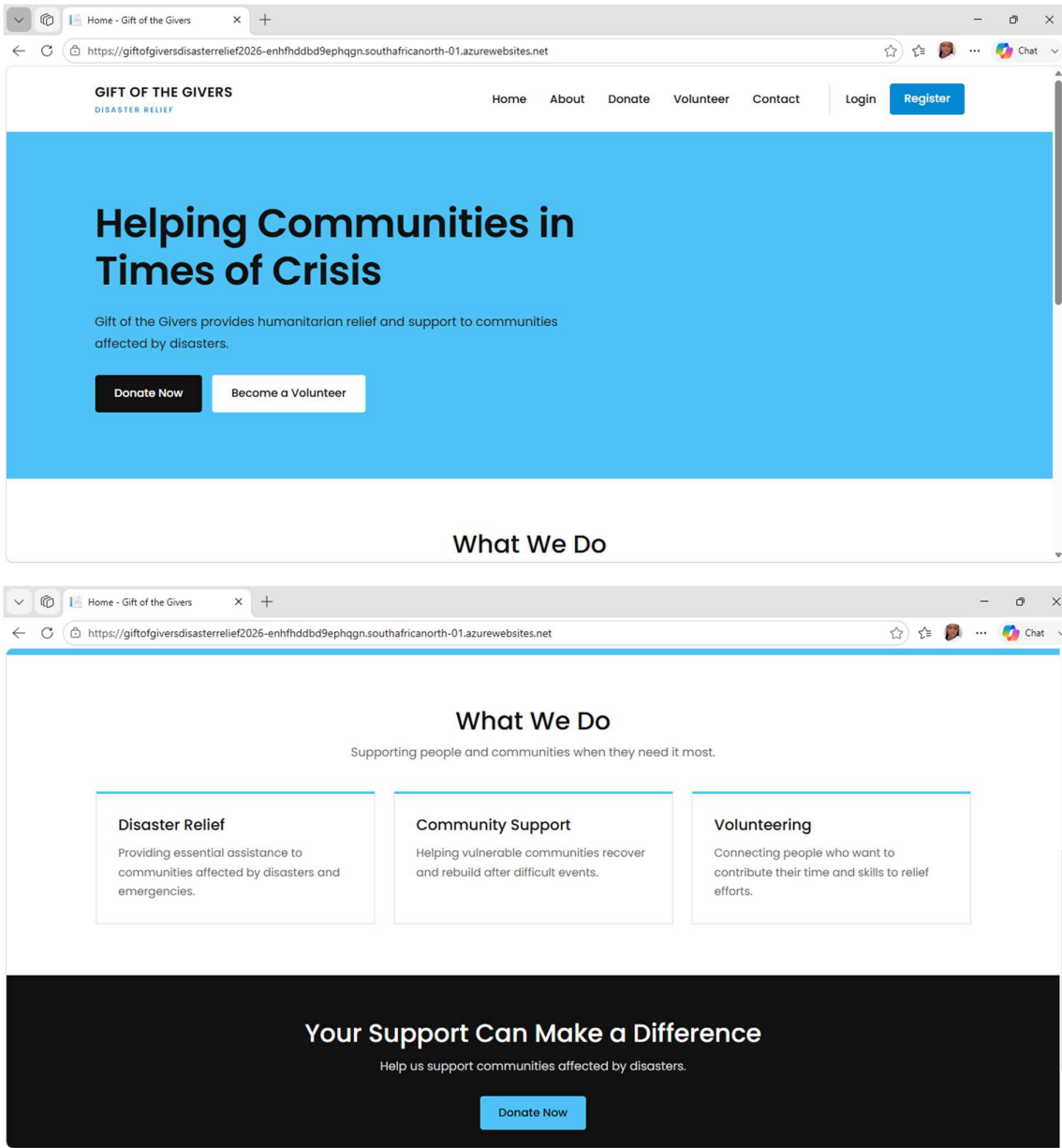


Figure 2: About the Gift of the Givers Disaster Relief System

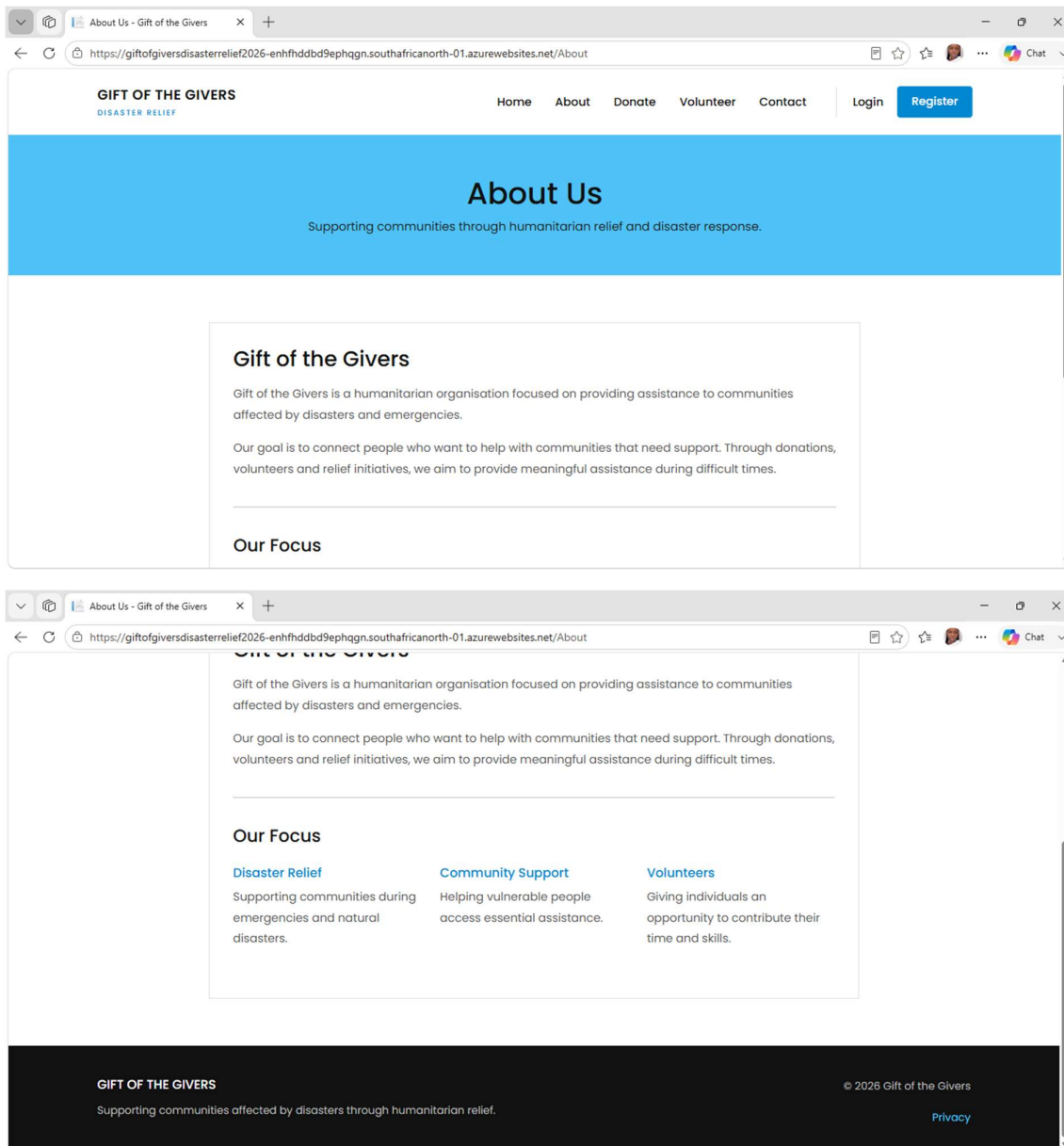


Figure 3: User Login Page

GIFT OF THE GIVERS
DISASTER RELIEF

Home About Donate Volunteer Contact Login Register

Login

Sign in to access your Gift of the Givers account.

Sign In

Select your account type and enter your login details.

Select Role

-- Select your role --

Email Address

Enter your email address

Password

Enter your password

☐ Remember me

Login

[Forgot your password?](#)

Don't have a donor account? [Register here](#)

GIFT OF THE GIVERS
Supporting communities affected by disasters through humanitarian relief.

© 2026 Gift of the Givers
[Privacy](#)

Figure 4: User Registration Page

Register - Gift of the Givers

https://giftogiversdisasterrelief2026-enhfhd9epqgn.southafricanorth-01.azurewebsites.net/Identity/Account/Register

GIFT OF THE GIVERS
DISASTER RELIEF

Home About Donate Volunteer Contact Login Register

Create Your Account

Register as a donor or employee to access the appropriate panel.

Create Account

Complete the form below to create your account.

Account Type

-- Select Account Type --

Select the type of account you want to create.

Full Name

Enter your full name

Email Address

Enter your email address

Password

Create a password

Confirm Password

Confirm your password

Create Account

Already have an account? [Login](#)

GIFT OF THE GIVERS
Supporting communities affected by disasters through humanitarian relief.

© 2026 Gift of the Givers
[Privacy](#)

Figure 5: Forgot Password Page

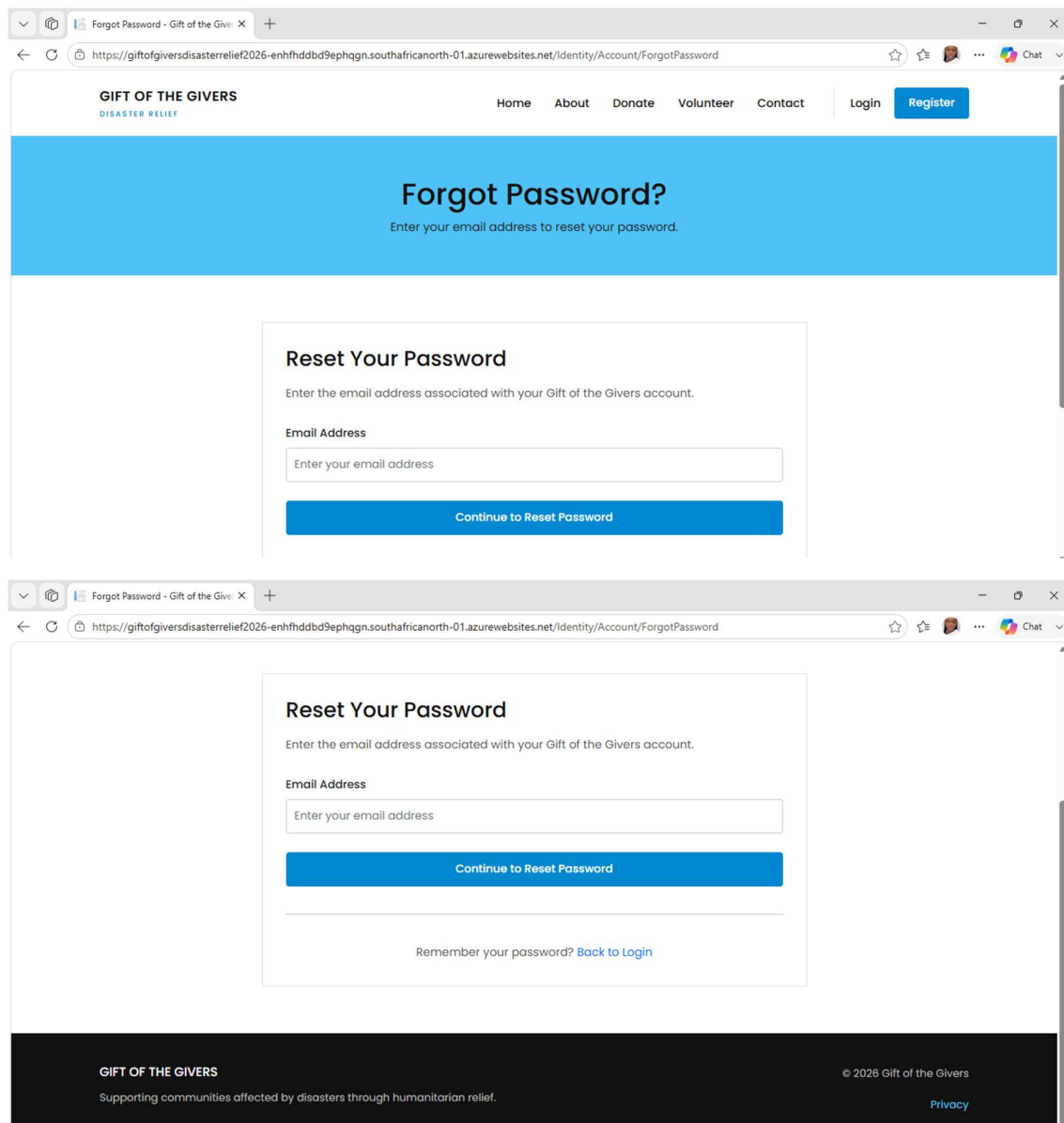


Figure 6: Reset Password Page

GIFT OF THE GIVERS
DISASTER RELIEF

Home About Donate Volunteer Contact Login Register

Reset Password

Create a new password for your account.

Reset Your Password

Enter your email address and create a new password below.

Email Address

New Password

Confirm New Password

Password Requirements

- Use at least 6 characters.
- Use a combination of letters and numbers.
- Make sure both passwords match.

Reset Password

Remember your password? [Back to Login](#)

GIFT OF THE GIVERS
Supporting communities affected by disasters through humanitarian relief.

© 2026 Gift of the Givers
[Privacy](#)

Figure 7: Public Donation Options Page

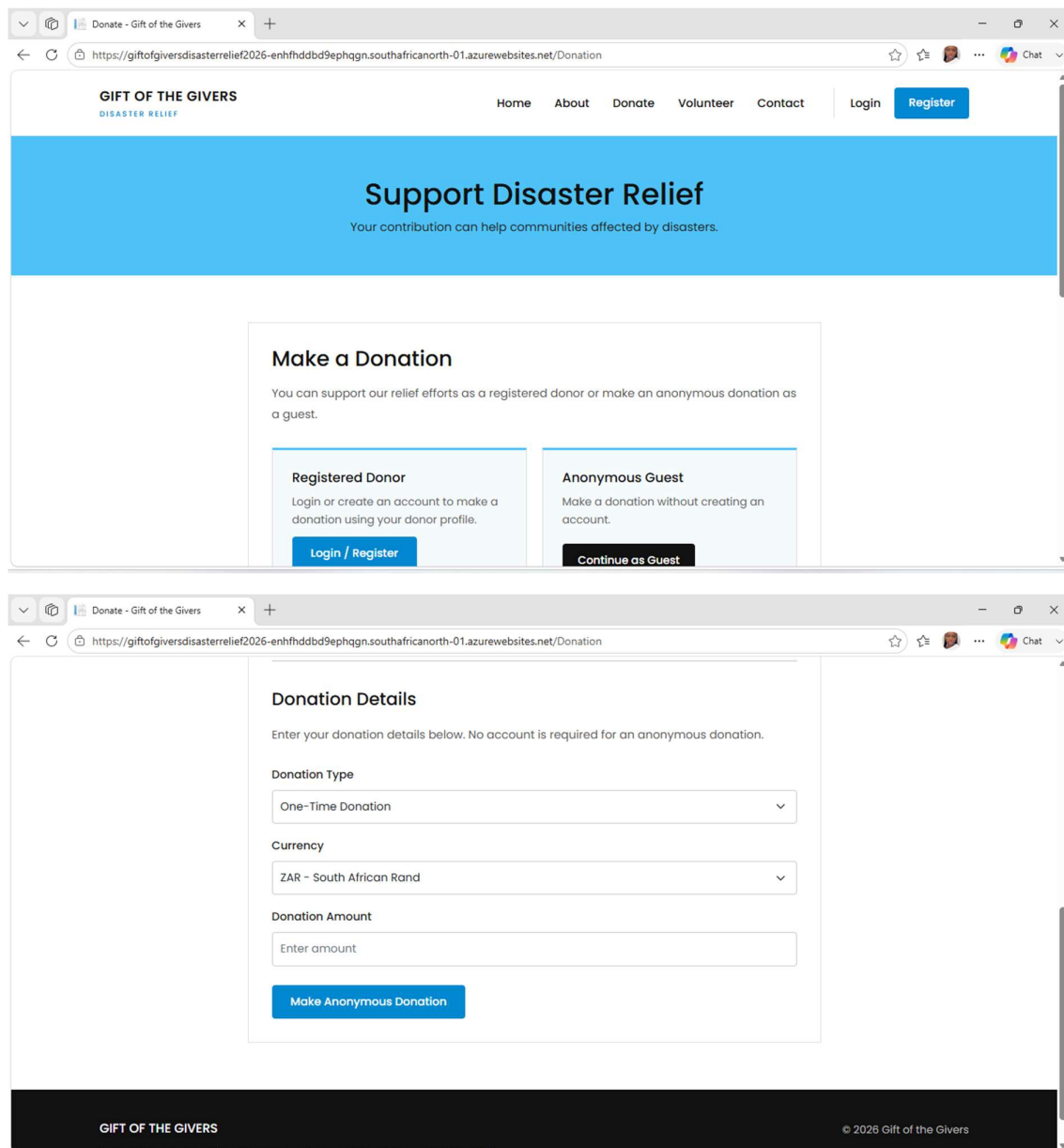


Figure 8: Donation Confirmation Page

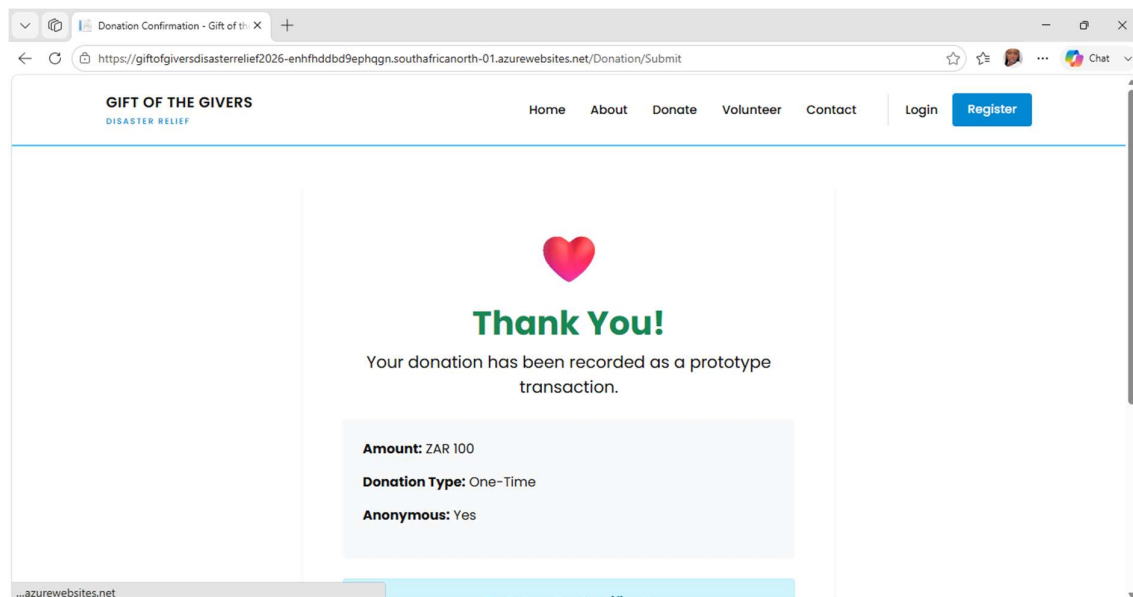
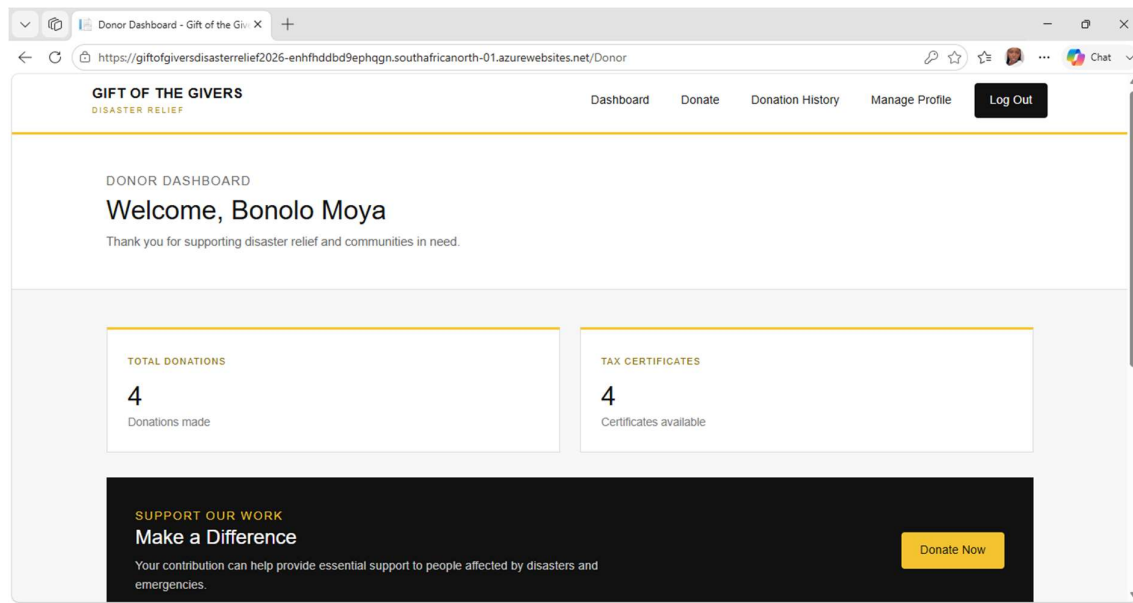


Figure 9: Donor Dashboard



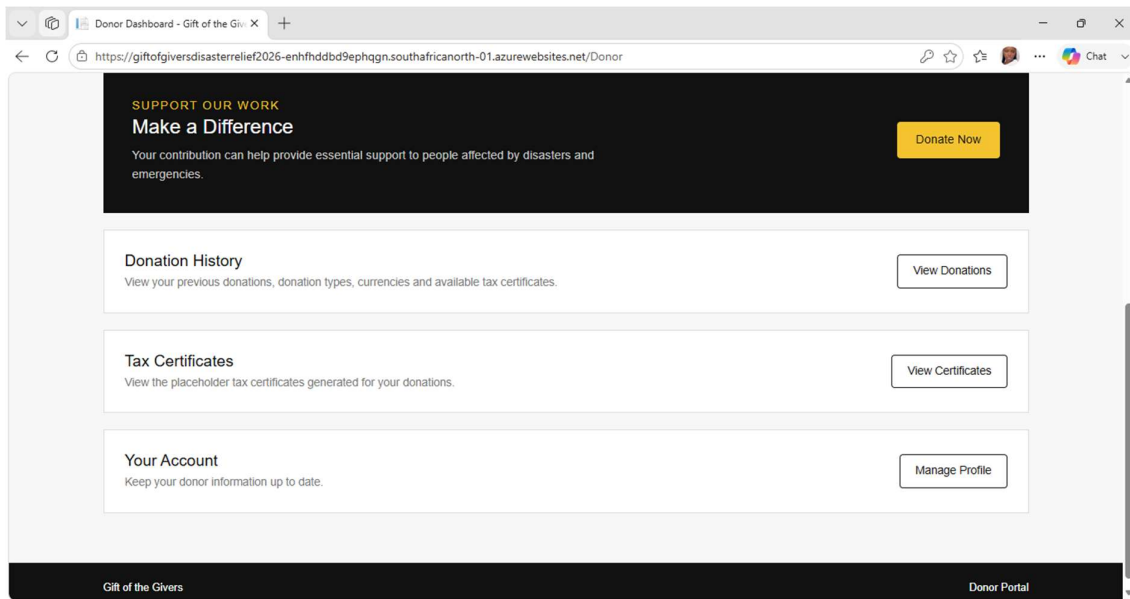


Figure 10: Donor Donation Form

The screenshot shows a web browser window with the title 'Donate - Gift of the Givers'. The URL is 'https://giftogiversdisasterrelief2026-enhfddbd9ephqgn.southafricanorth-01.azurewebsites.net/Donor/Donate'. The page has a header with 'GIFT OF THE GIVERS' and 'DISASTER RELIEF'. The main heading is 'DONATION Make a Difference'. Below this is a 'Make a Donation' section with a 'Choose how you would like to support Gift of the Givers.' instruction. The form includes dropdowns for 'Donation Type' (One-Time Donation), 'Currency' (ZAR - South African Rand), and a 'Donation Amount' input field with the value '1000'.

Make a Donation

Choose how you would like to support Gift of the Givers.

Donation Type

One-Time Donation

Currency

ZAR - South African Rand

Donation Amount

100

Donation Information

This is a prototype donation system. No real payment will be processed. A placeholder tax certificate will be generated after the donation is recorded.

[Continue Donation](#) [Back to Dashboard](#)

Gift of the Givers Donor Portal

Figure 11: Donor Donation History Page

GIFT OF THE GIVERS
DISASTER RELIEF

Dashboard Donate Donation History Manage Profile Log Out

DONATION SUCCESSFUL

Thank You for Your Donation

Your support helps provide essential assistance to communities affected by disasters and emergencies.

Donation Recorded Successfully

Your donation has been recorded successfully.

Donation Details

Donation Amount	ZAR 100.00
Donation Type	One-Time
Currency	ZAR
Donation Date	27 August 2026, 23:46

...azurewebsites.net/Donor

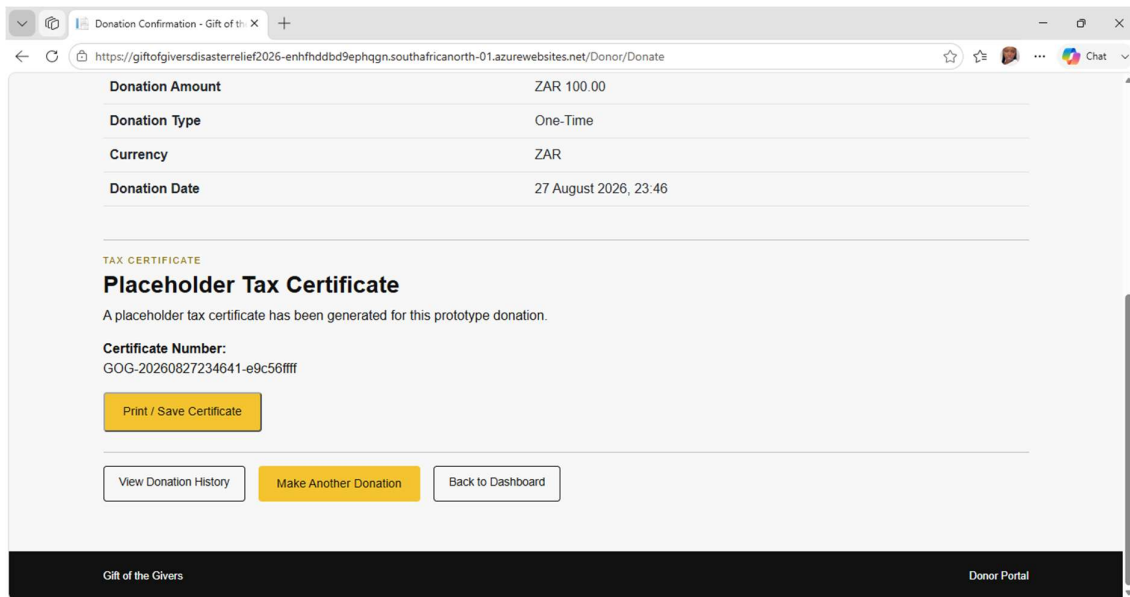


Figure 12: Tax Certificate Information Page

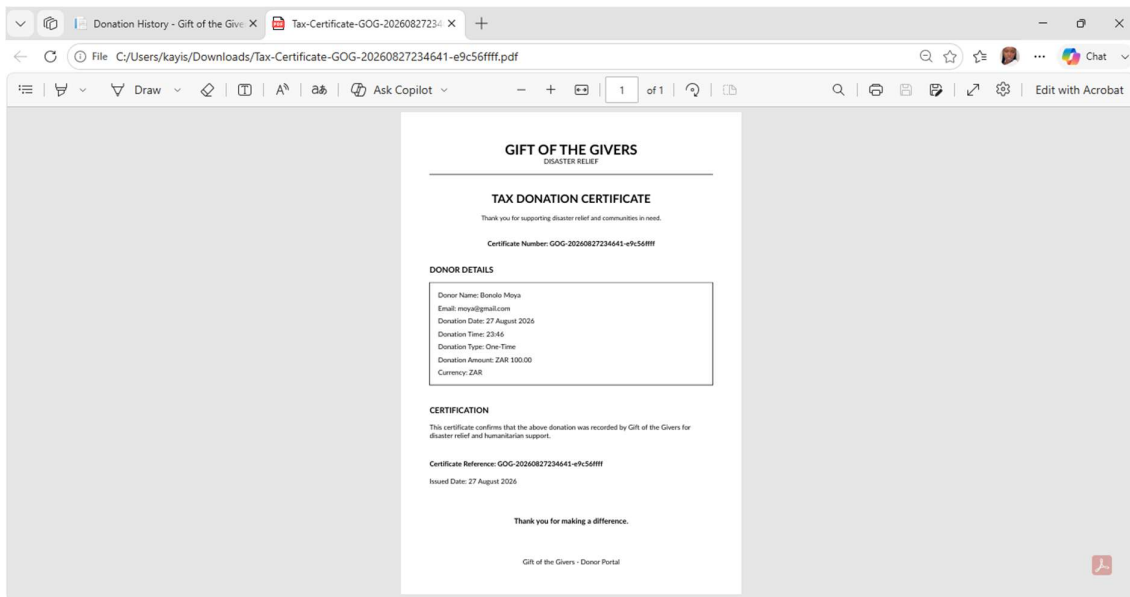


Figure 13: Volunteer Registration Page

The image shows two screenshots of a web browser displaying the 'Become a Volunteer' page for 'GIFT OF THE GIVERS DISASTER RELIEF'. The browser's address bar shows the URL: `https://giftogiversdisasterrelief2026-enhfddbd9ephqgn.southafricanorth-01.azurewebsites.net/Volunteer`. The page has a navigation bar with links: Home, About, Donate, Volunteer, Contact, Login, and a Register button. The main heading is 'Become a Volunteer' with the subtext 'Give your time and skills to help communities in need.' Below this is a 'Volunteer Interest Form' with the instruction 'Tell us a little about yourself and how you would like to contribute.' The form contains the following fields: 'Full Name' (text input), 'Email Address' (text input), 'Skills' (text input with placeholder 'e.g. First Aid, Cooking, Logistics'), and 'Availability' (dropdown menu with 'Select your availability'). A blue 'Submit Interest' button is at the bottom of the form. The footer of the page includes the 'GIFT OF THE GIVERS' logo, the tagline 'Supporting communities affected by disasters through humanitarian relief.', the copyright notice '© 2026 Gift of the Givers', and a 'Privacy' link.

GIFT OF THE GIVERS
DISASTER RELIEF

Home About Donate Volunteer Contact Login [Register](#)

Become a Volunteer

Give your time and skills to help communities in need.

Volunteer Interest Form

Tell us a little about yourself and how you would like to contribute.

Full Name

Email Address

Skills

Availability

[Submit Interest](#)

GIFT OF THE GIVERS
Supporting communities affected by disasters through humanitarian relief.

© 2026 Gift of the Givers [Privacy](#)

Figure 14: Employee Dashboard

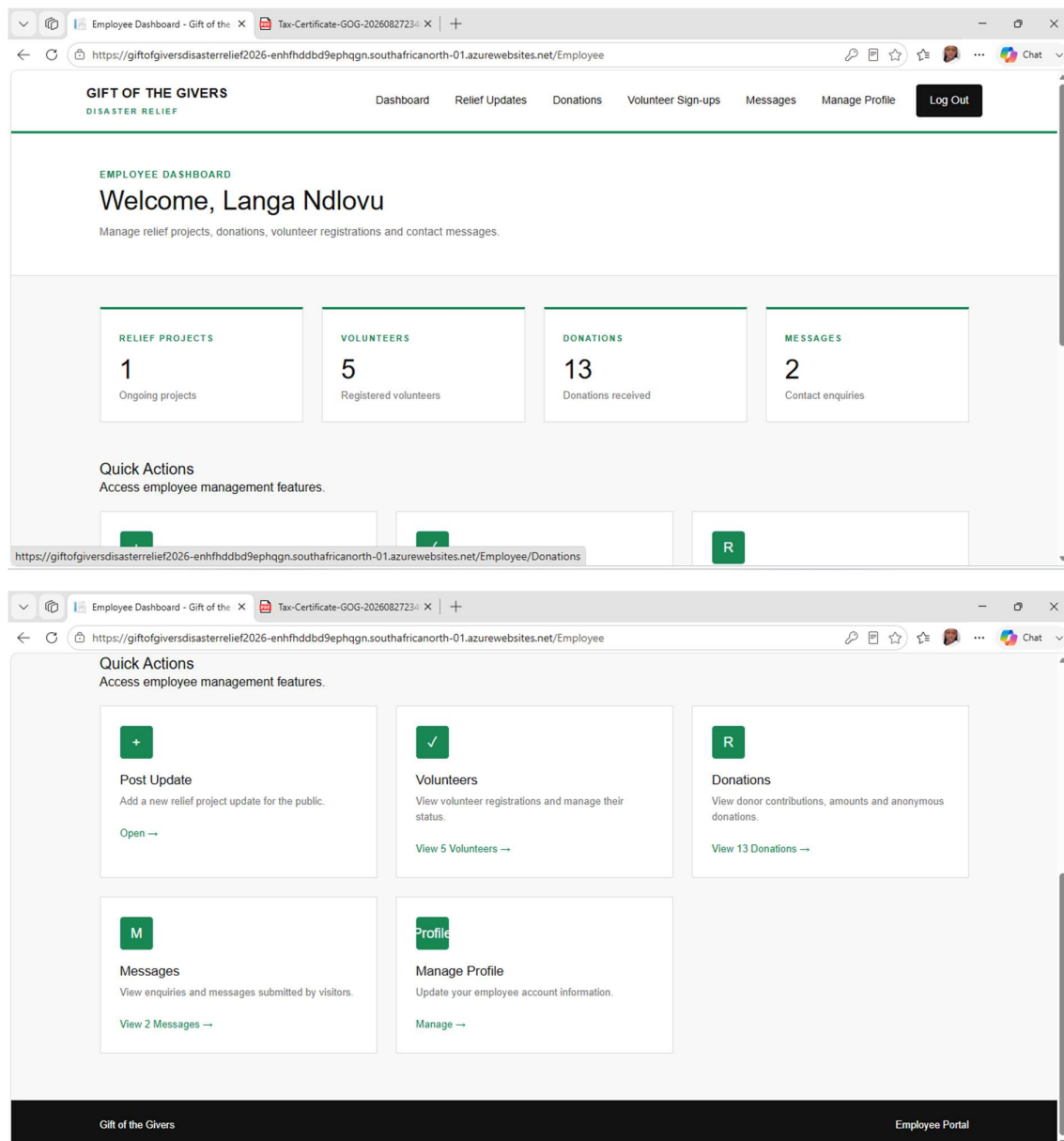


Figure 15: Employee Donation Management Page

Donations - Gift of the Givers
Tax-Certificate-GOG-20260827234

https://giftogiversdisasterrelief2026-enhfhdbd9ephqgn.southafricanorth-01.azurewebsites.net/Employee/Donations

GIFT OF THE GIVERS
DISASTER RELIEF

Dashboard
Relief Updates
Donations
Volunteer Sign-ups
Messages
Manage Profile
Log Out

DONATION MANAGEMENT

Donor Donations

View donations received from donors and monitor contributions to disaster relief.

TOTAL DONATIONS
13
Donations received

IDENTIFIED DONORS
8
Donor information available

ANONYMOUS DONATIONS
5
Donor identity hidden

CONTRIBUTIONS

All Donations

Employees can view donor details, donation amounts and anonymous contributions.

Donations - Gift of the Givers
Tax-Certificate-GOG-20260827234

https://giftogiversdisasterrelief2026-enhfhdbd9ephqgn.southafricanorth-01.azurewebsites.net/Employee/Donations

Figure 16: Employee Volunteer Management Page

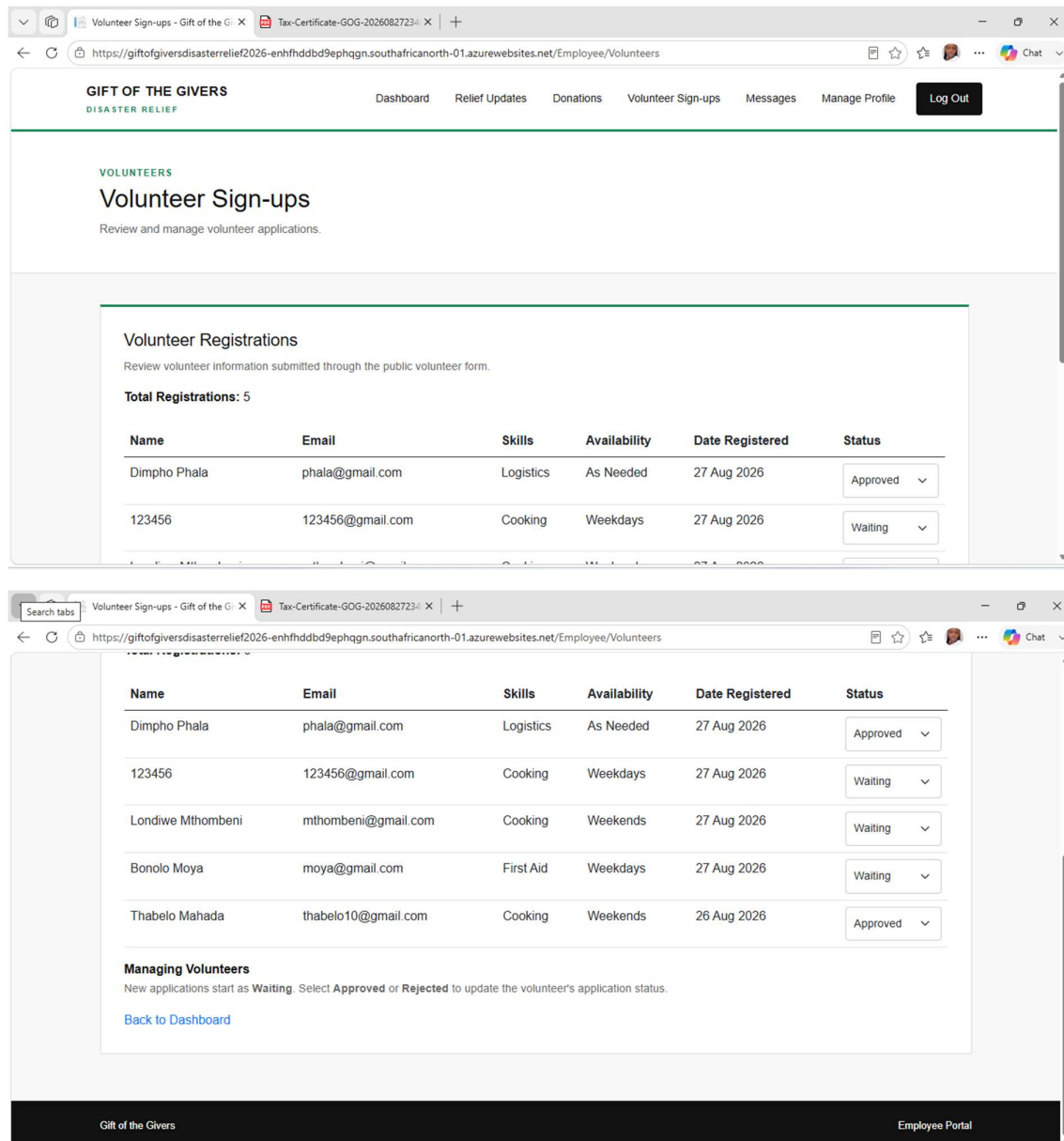


Figure 17: Employee Relief Project Updates Page

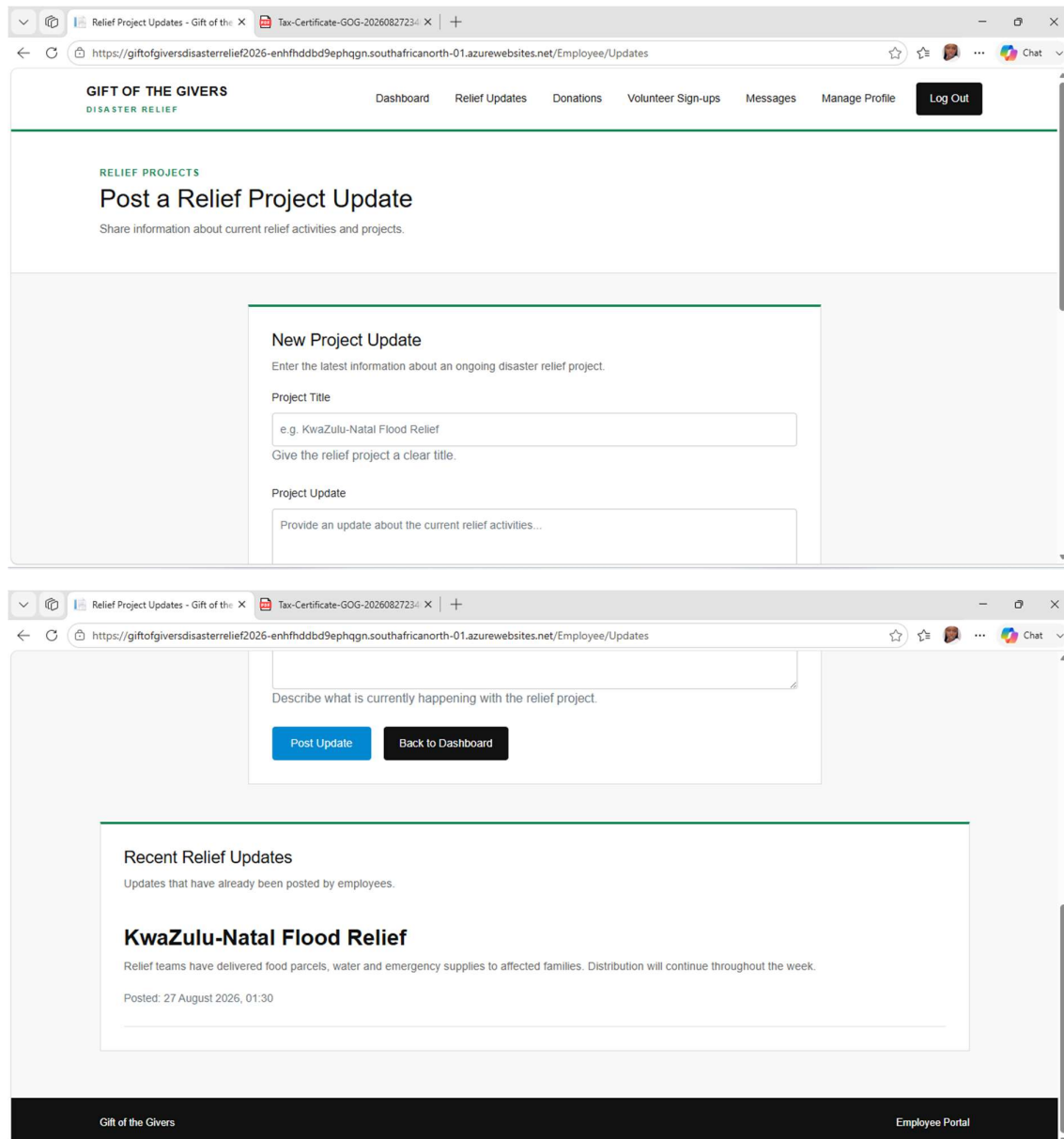


Figure 18: Employee Profile Management Page

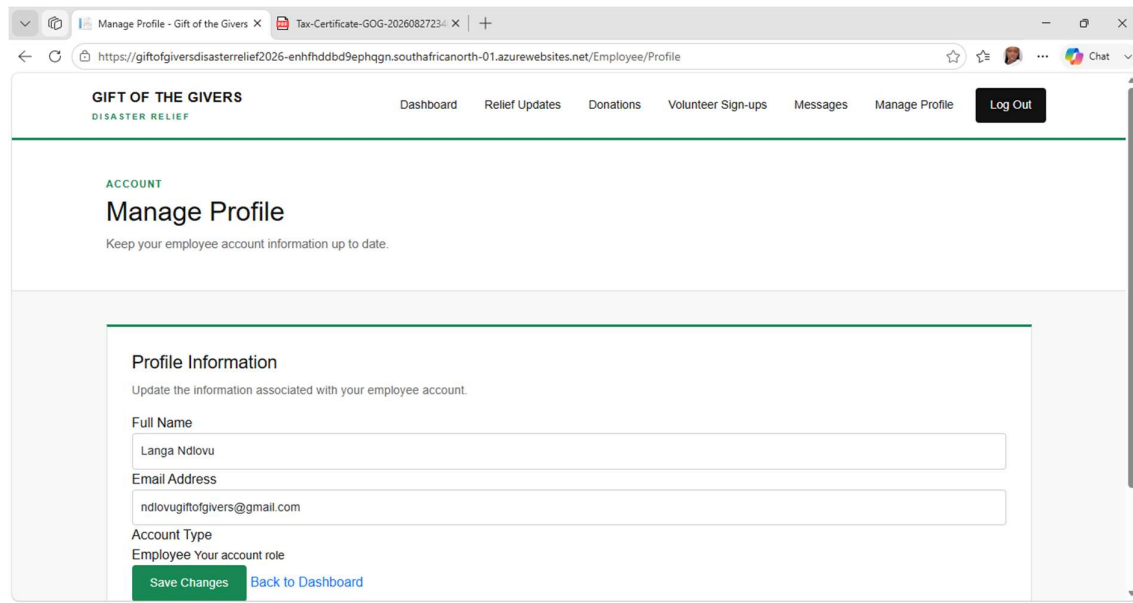


Figure 19: Employee Messages Page from Contact Page

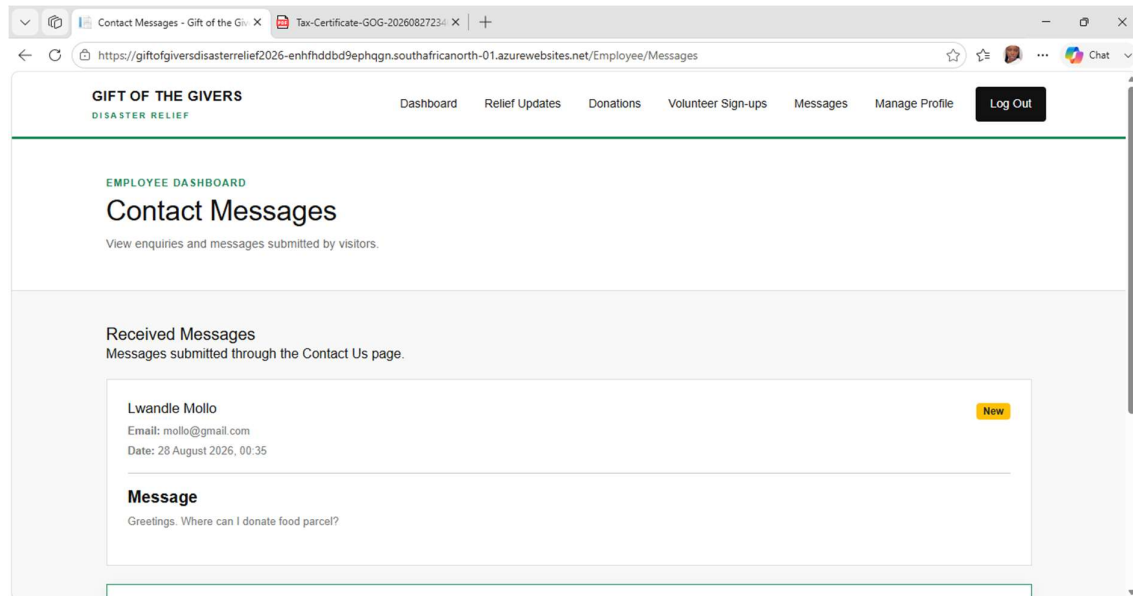
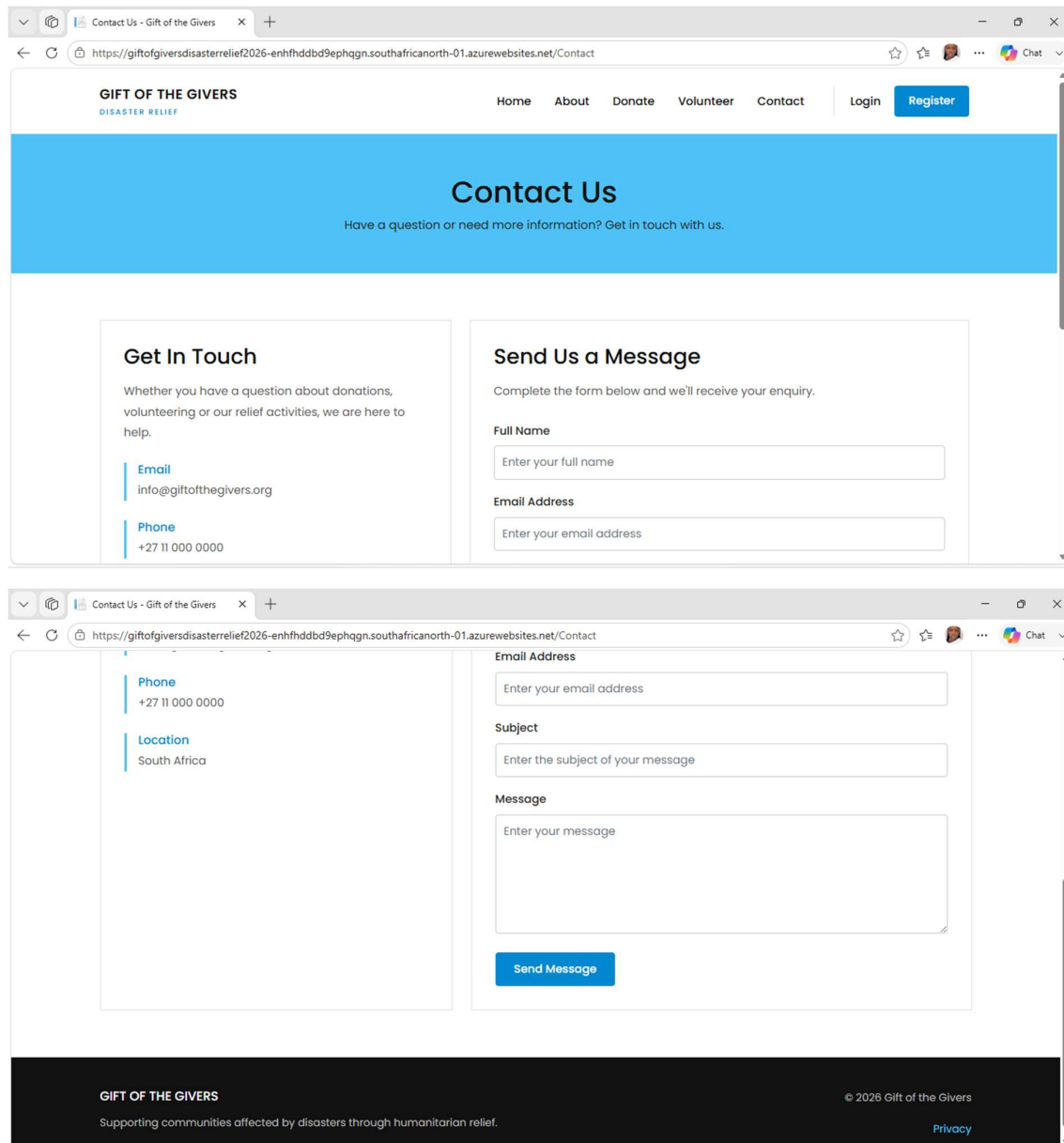


Figure 20: Contact Page



3.2.2 Group Contribution Reflections

Thabelo Mahada – User Interface (UI) Design

This group member focused on designing and developing the user interface of the web application. They worked on the layout, navigation, forms, buttons, colours, and overall appearance of the different pages to ensure that the system was easy to use and visually consistent.

Kgolagano Mphai – Authentication and User Management

This group member was responsible for implementing authentication and user management. They worked on the login, registration, logout, forgot password, and reset password functionality. They also assisted with role-based access for donors and employees.

Nkateko Mabasa – Donation Functionality

This group member focused on the donation functionality of the system. They worked on the public donation page, donor donation process, donation confirmation, donation history, anonymous donations, and displaying donation information to employees.

Thubalakhe Lungelo Mthembu – Volunteer and Employee Functionality

This group member worked on the volunteer and employee features of the application. They developed the volunteer registration functionality and assisted with the employee dashboard, volunteer management, relief project updates, and employee profile functionality.

3.3 Publishing to Azure App Services

3.3.1 Provide the live URL of your deployed app.

<https://giftofgiversdisasterrelief2026-enhfhddbd9ephqgn.southafricanorth-01.azurewebsites.net>

3.3.2 Include screenshots of the deployment process.

Figure X: Azure App Service created for the Gift of the Givers Disaster Relief System.

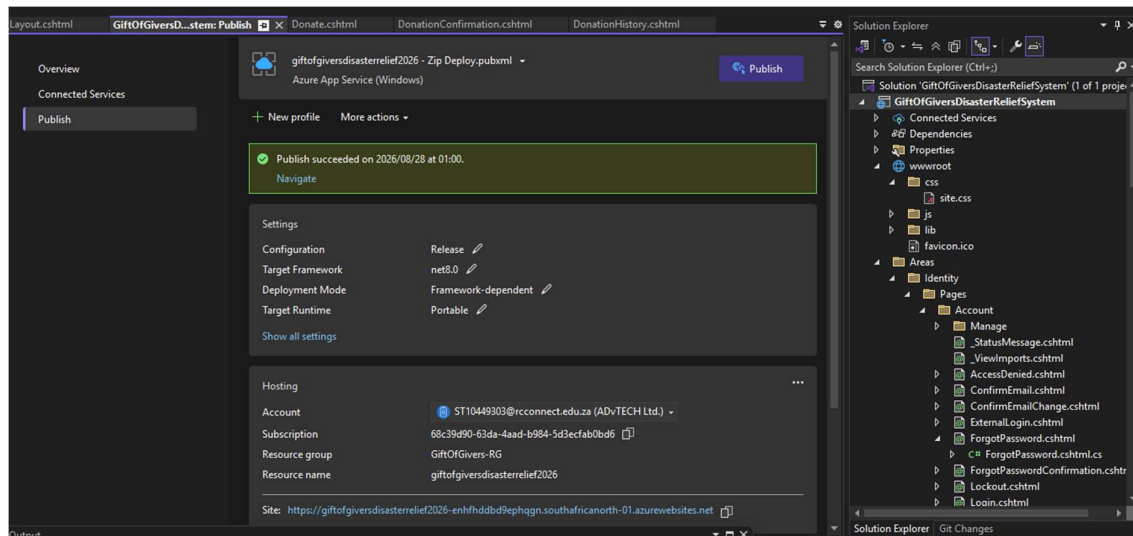


Figure X: Successful deployment of the Gift of the Givers Disaster Relief System using Visual Studio Zip Deploy.

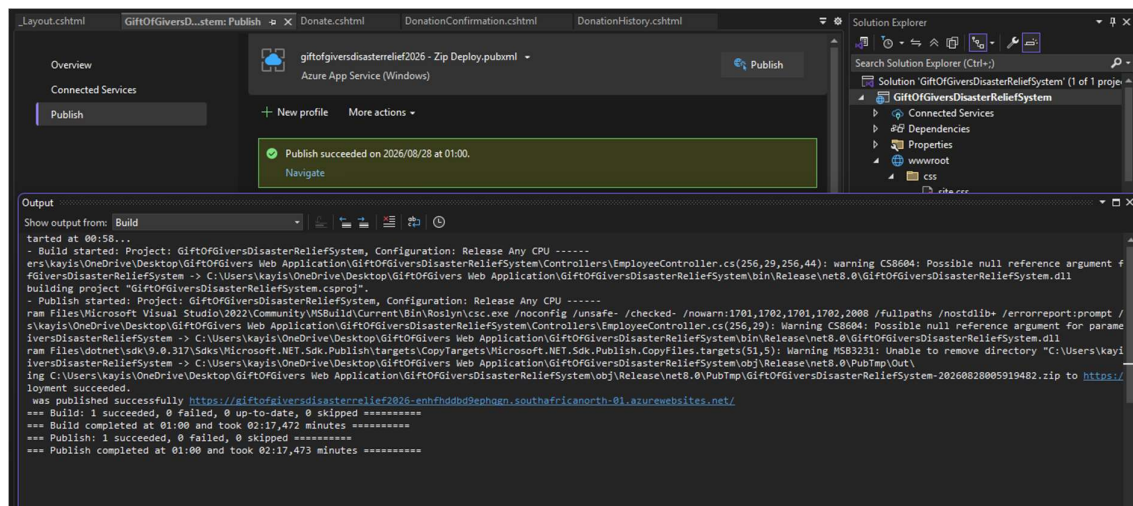
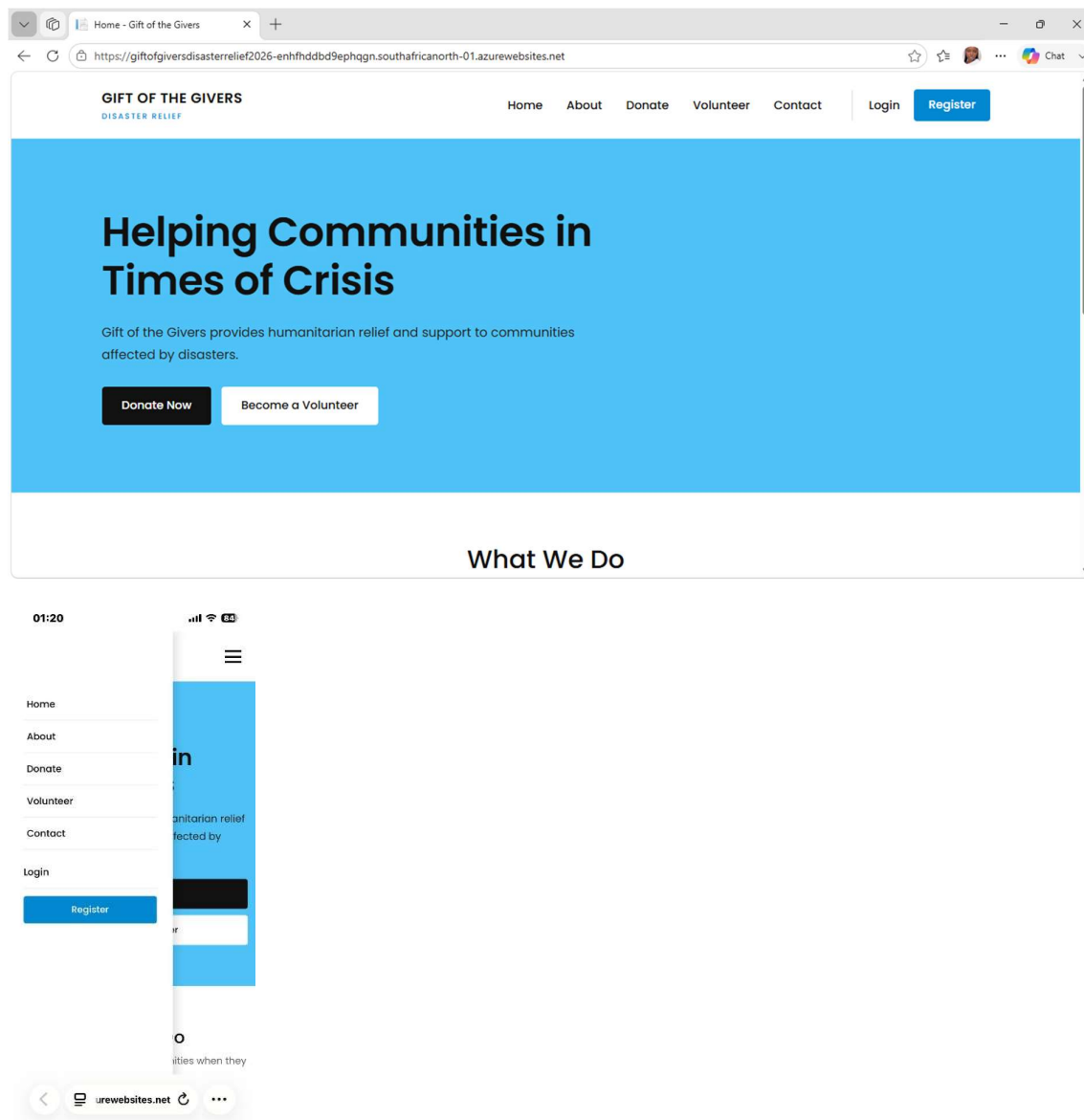


Figure X: Live Gift of the Givers Disaster Relief System running on Microsoft Azure.



3.3.3 Reflect on why cloud deployment is vital for this project (e.g., real-time scalability during disaster events).

Cloud deployment is important because it allows the system to be accessed from anywhere and by many users during disasters. Microsoft Azure provides scalability and availability, allowing the system to handle increased traffic when more people need to make donations, register as volunteers or view relief updates. It also allows employees to update information quickly without users needing to reinstall the application. Overall, cloud deployment makes the system accessible, flexible and reliable during emergencies (GeeksForGeeks, 2026).

REFERENCES

Live URL of deployed app.

[Home - Gift of the Givers](#)

<https://giftofgiversdisasterrelief2026-enhfhddbd9ephqgn.southafricanorth-01.azurewebsites.net>

Azure Boards URL:

<https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/boards/board/t/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System%20Team/Stories>

<https://dev.azure.com/ST10449303/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System/backlogs/backlog/Gift%20Of%20The%20Givers%20Disaster%20Relief%20System%20Team/Stories>

GeeksForGeeks, 2026. *Cloud Deployment Models*. [Online] Available at:

<https://www.geeksforgeeks.org/computer-science-fundamentals/cloud-deployment-models/> [Accessed 02 March 2026].

GeeksForGeeks, 2023. *Differences Between .NET Core and .NET Framework*.

[Online] Available at: <https://www.geeksforgeeks.org/c-sharp/differences-between-net-core-and-net-framework/> [Accessed 17 July 2023].

Microsoft Learn, 2026. *Indexes – SQL Server*. [Online] Available at:

<https://learn.microsoft.com/en-us/sql/relational-databases/indexes/indexes?view=sql-server-ver17> [Accessed 20 July 2026].

GeeksForGeeks, 2026. *Database Design in DBMS*. [Online] Available at:

<https://www.geeksforgeeks.org/dbms/database-design-in-dbms/> [Accessed 1 May 2026].

Microsoft Ignite, 2026. *About work items and work item types*. [Online] Available at:

<https://learn.microsoft.com/en-us/azure/devops/boards/work-items/about-work-items?view=azure-devops&tabs=agile-process> [Accessed 17 June 2026].

